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(54) **AUTONOMOUS INJECTION FLOW CONTROL VALVE**

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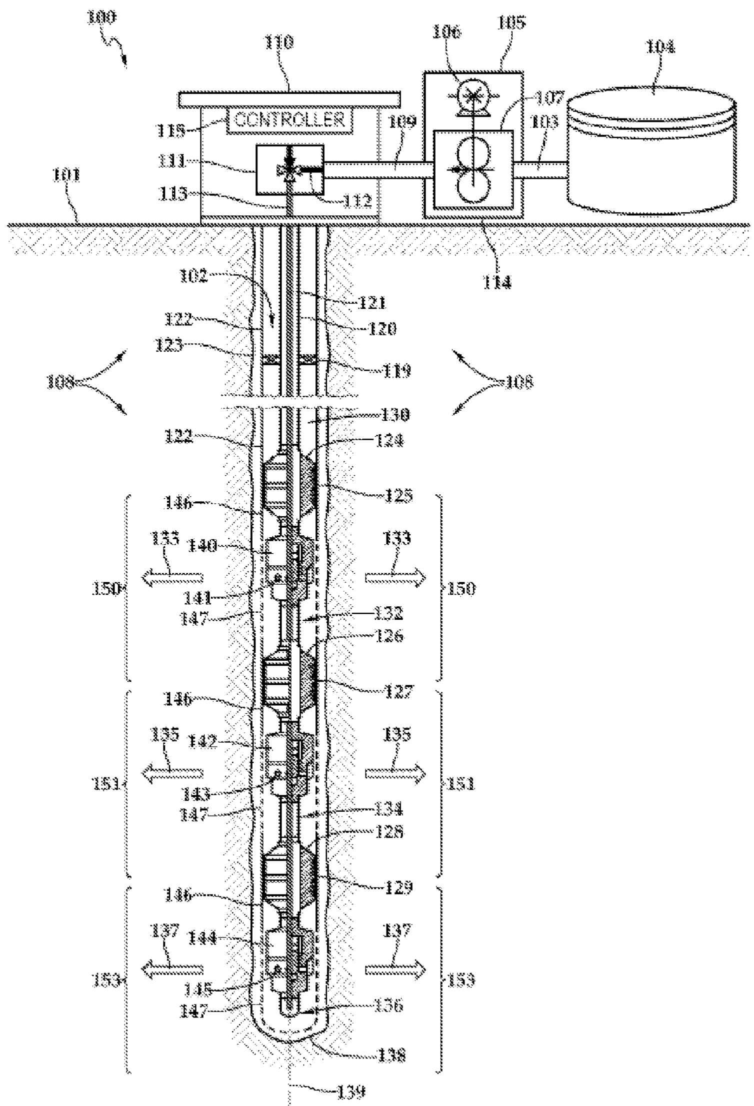
(52) **U.S. Cl.**

CPC ***E21B 34/10*** (2013.01); ***E21B 41/0064*** (2013.01); *F16K 5/0647* (2013.01)

(57) **ABSTRACT**

Provided is a flow control valve, a well system, and a method. The flow control valve, in one aspect, includes a tubular configured to be positioned downhole in a wellbore, the tubular having a central longitudinal axis, and a piston located in the tubular, the piston configured to separate injection fluid located within the tubular from a working fluid located in a sealed working fluid chamber. The flow control valve, according to this aspect, further includes a valve assembly located within the tubular and coupled with the piston, wherein the piston is configured to axially slide within the tubular to move the valve assembly between an open position that provides a fluid passageway for the injection fluid from the tubular into a subterranean formation and a closed position that closes the fluid passageway for the injection fluid from the tubular into the subterranean formation, based upon a density of the injection fluid.



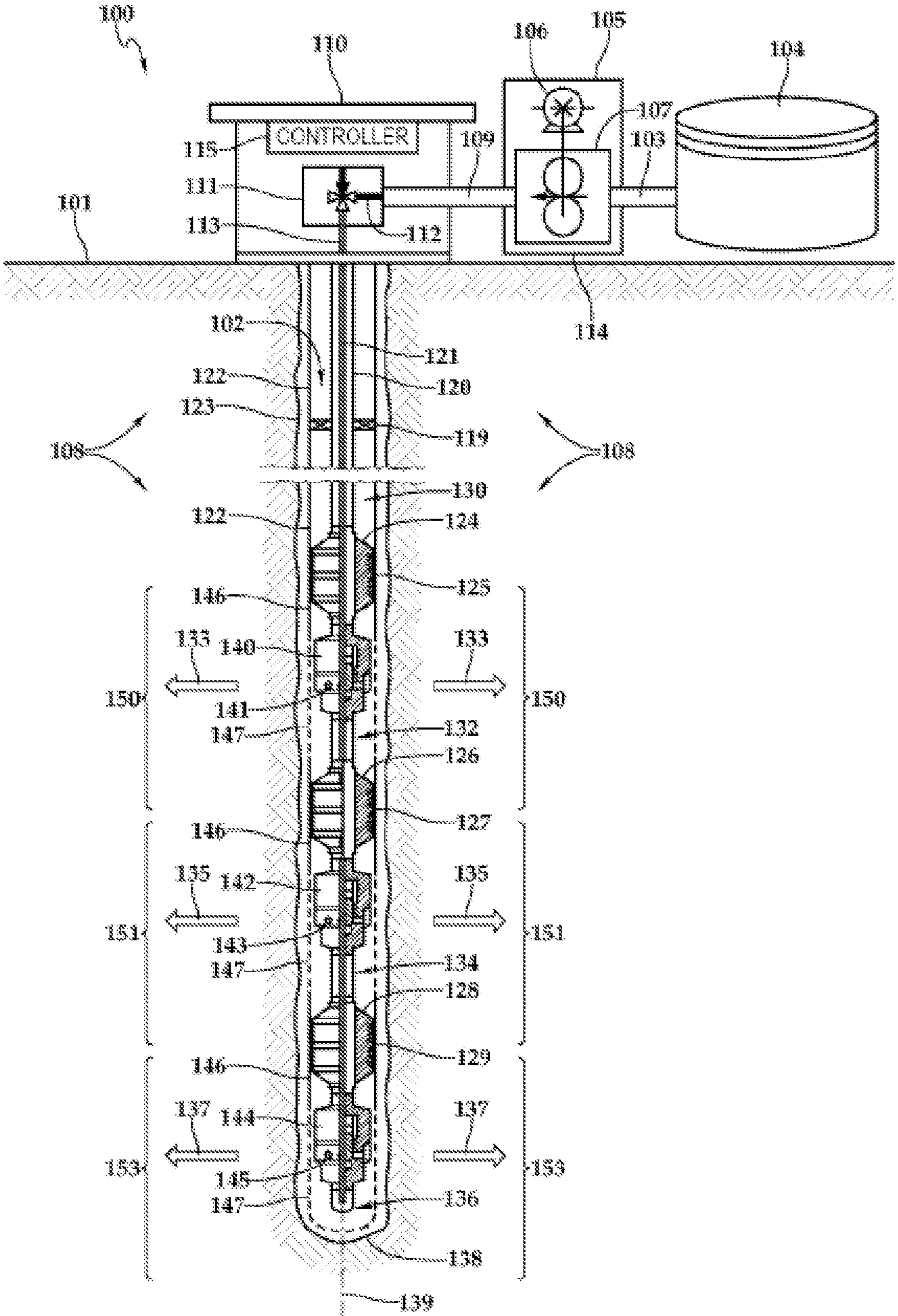

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FIG. 1A

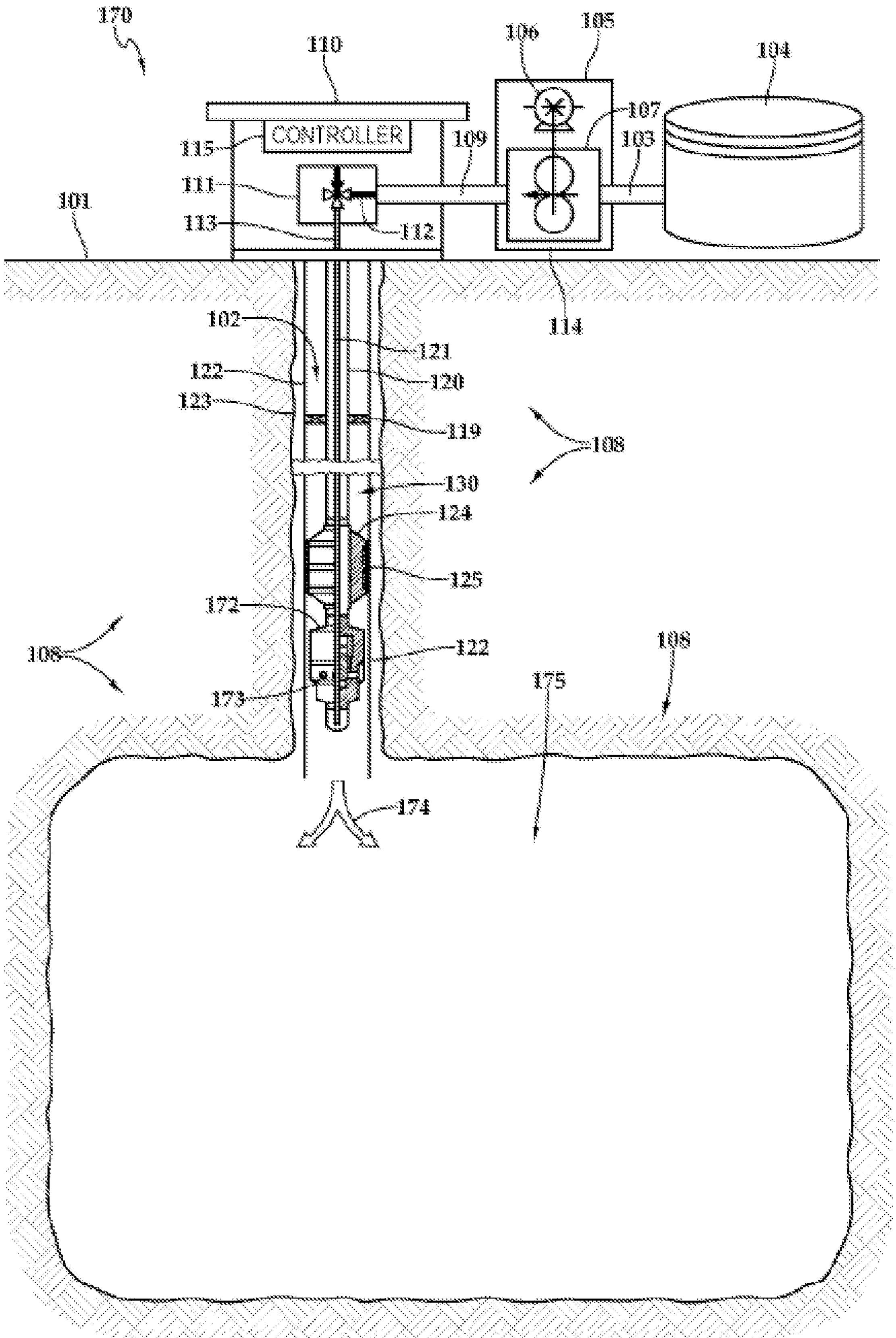
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FIG. 1B

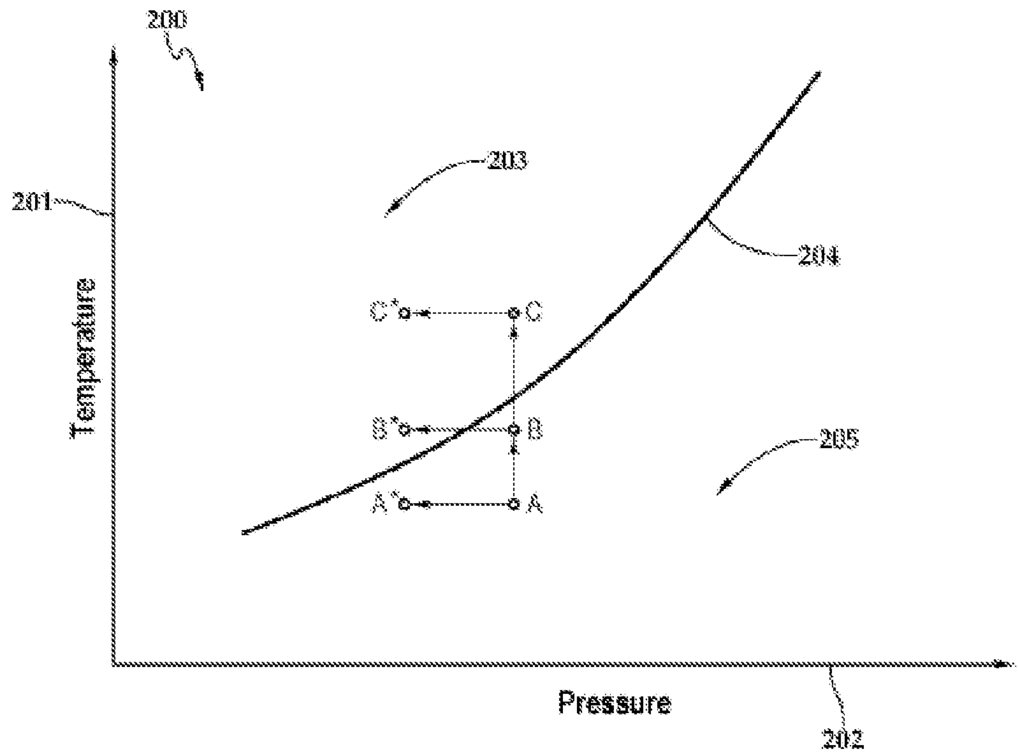


FIG. 2A

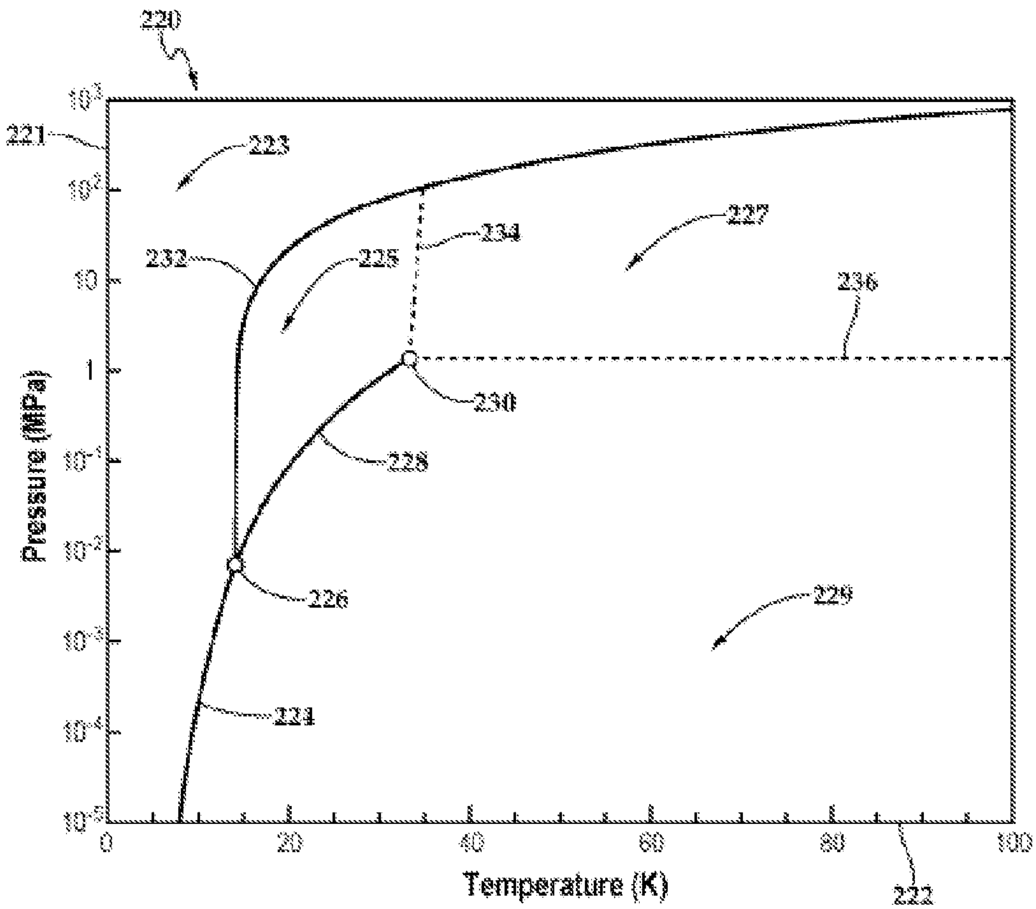

220
221
223
227
232
225
234
236
230
228
226
229
224
222
Pressure (MPa)
Temperature (K)
10^3
10^2
10
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
20
40
60
80
100

FIG. 2B

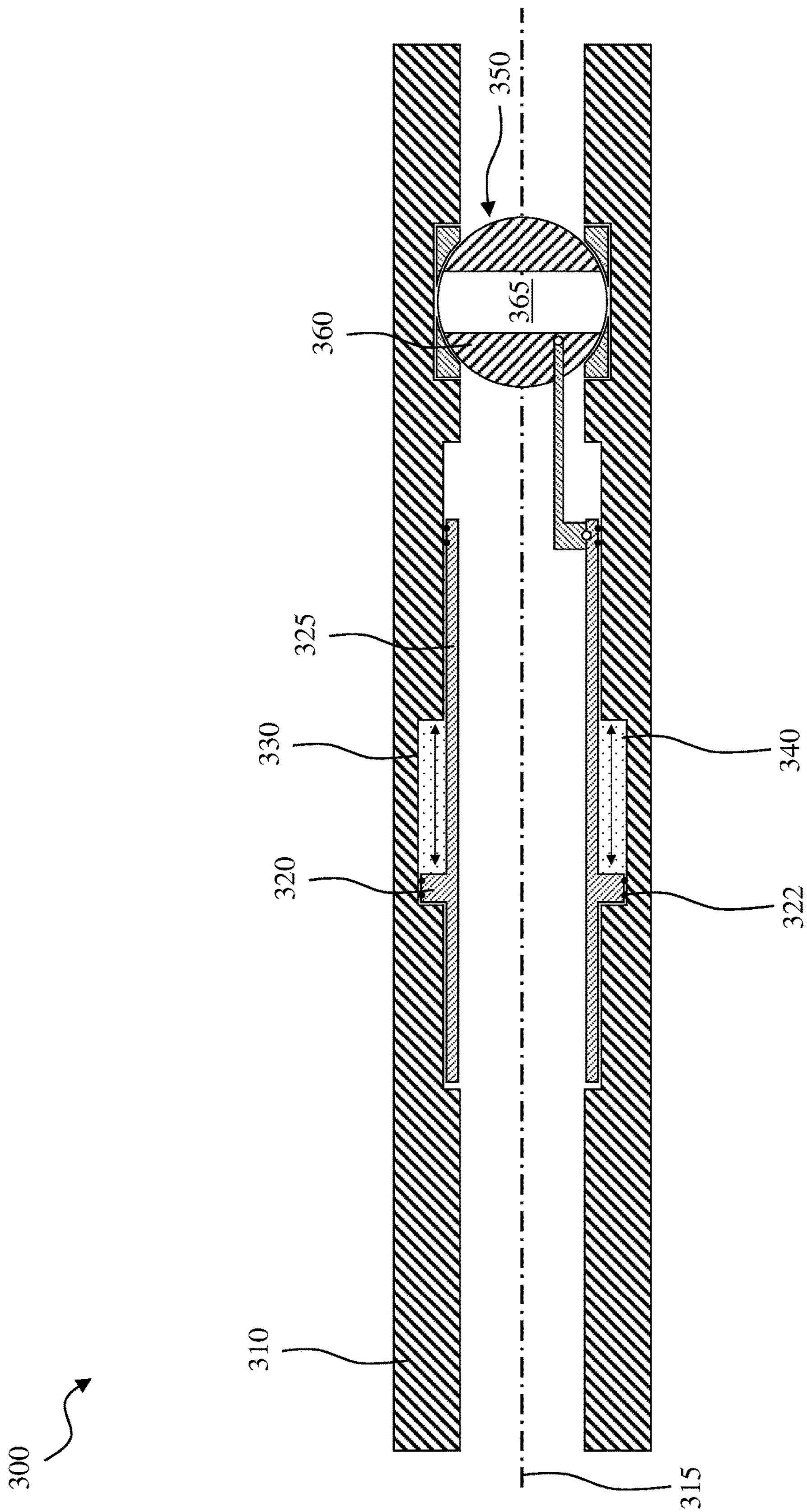


FIG. 3

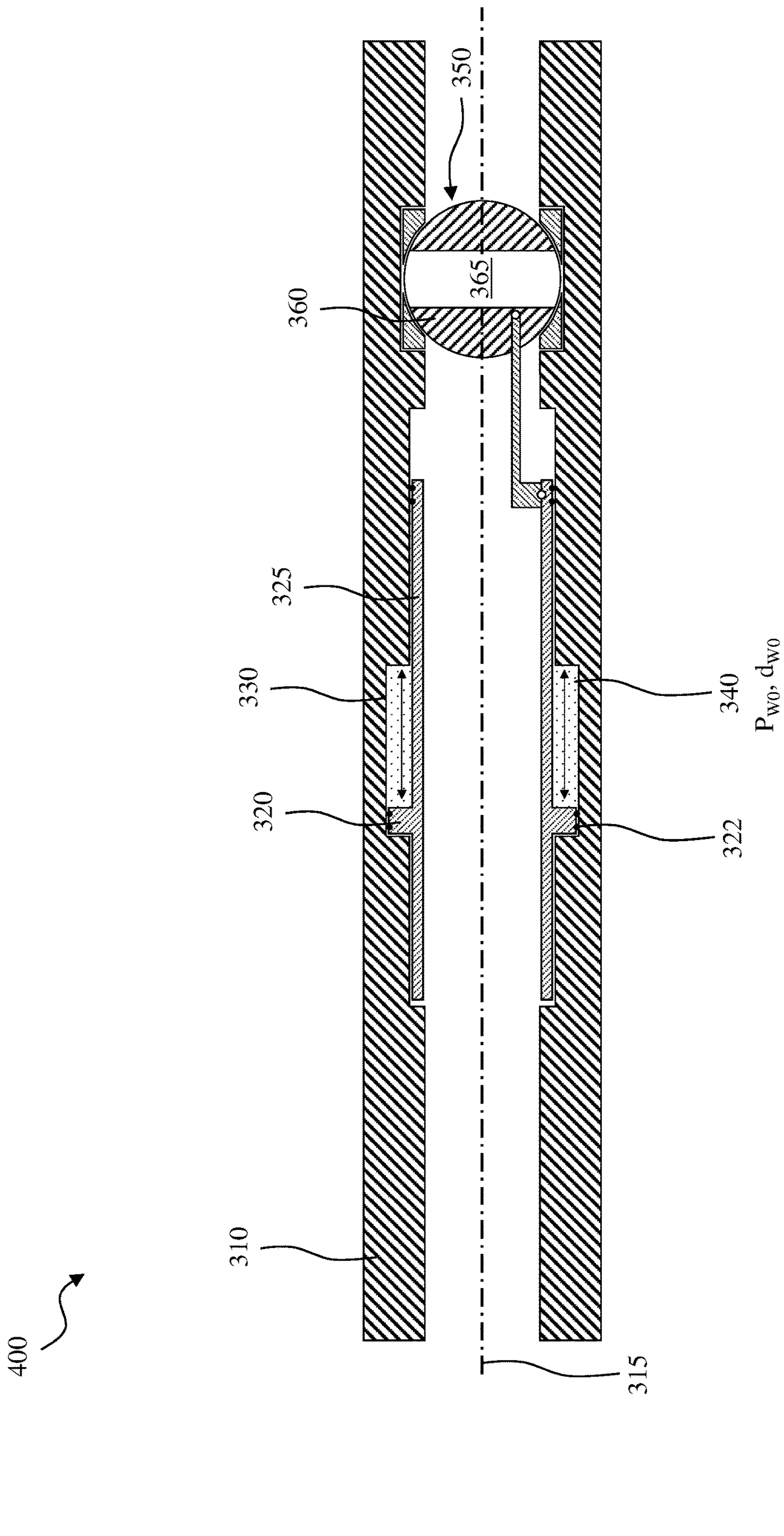


FIG. 4A

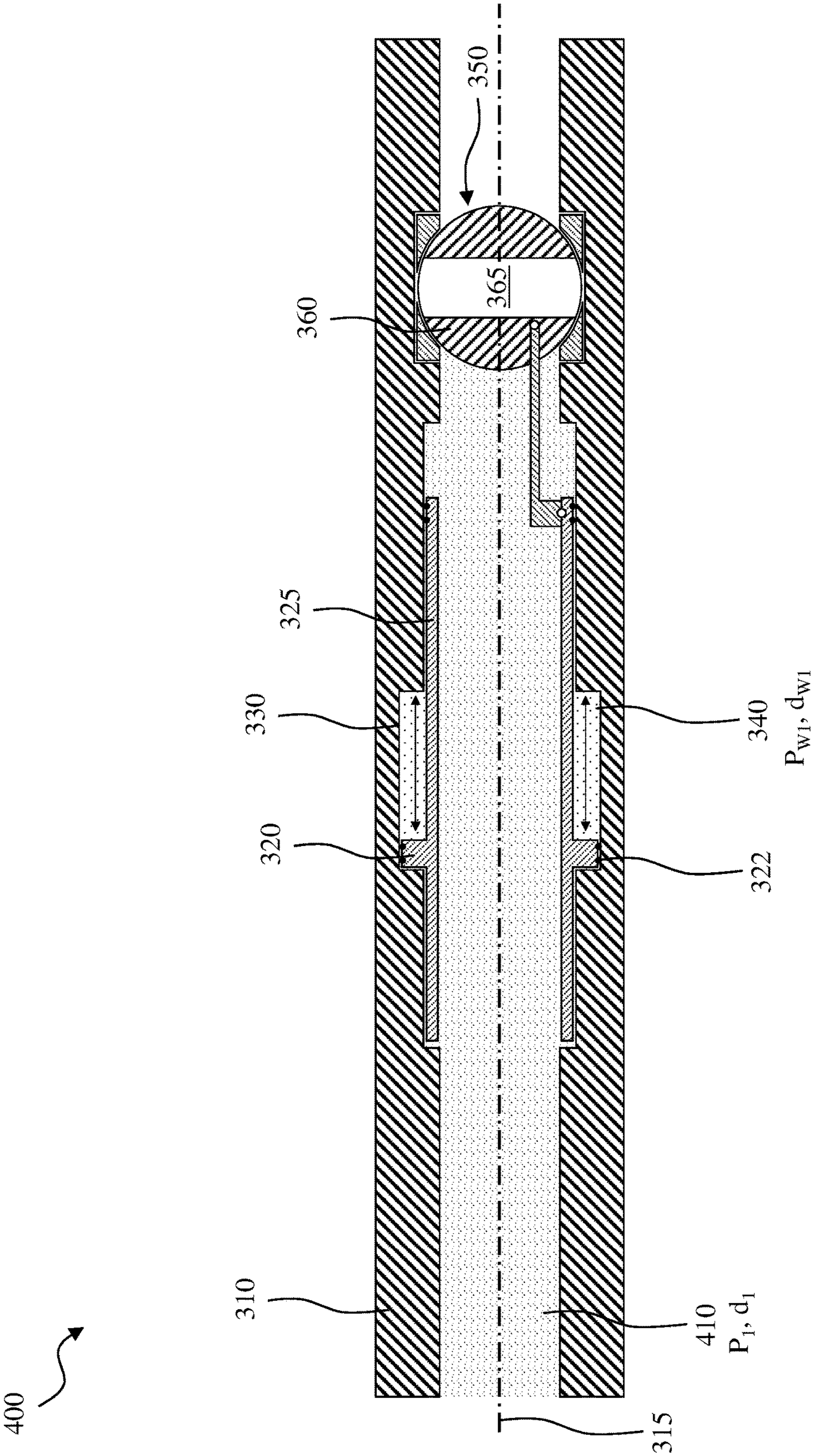


FIG. 4B

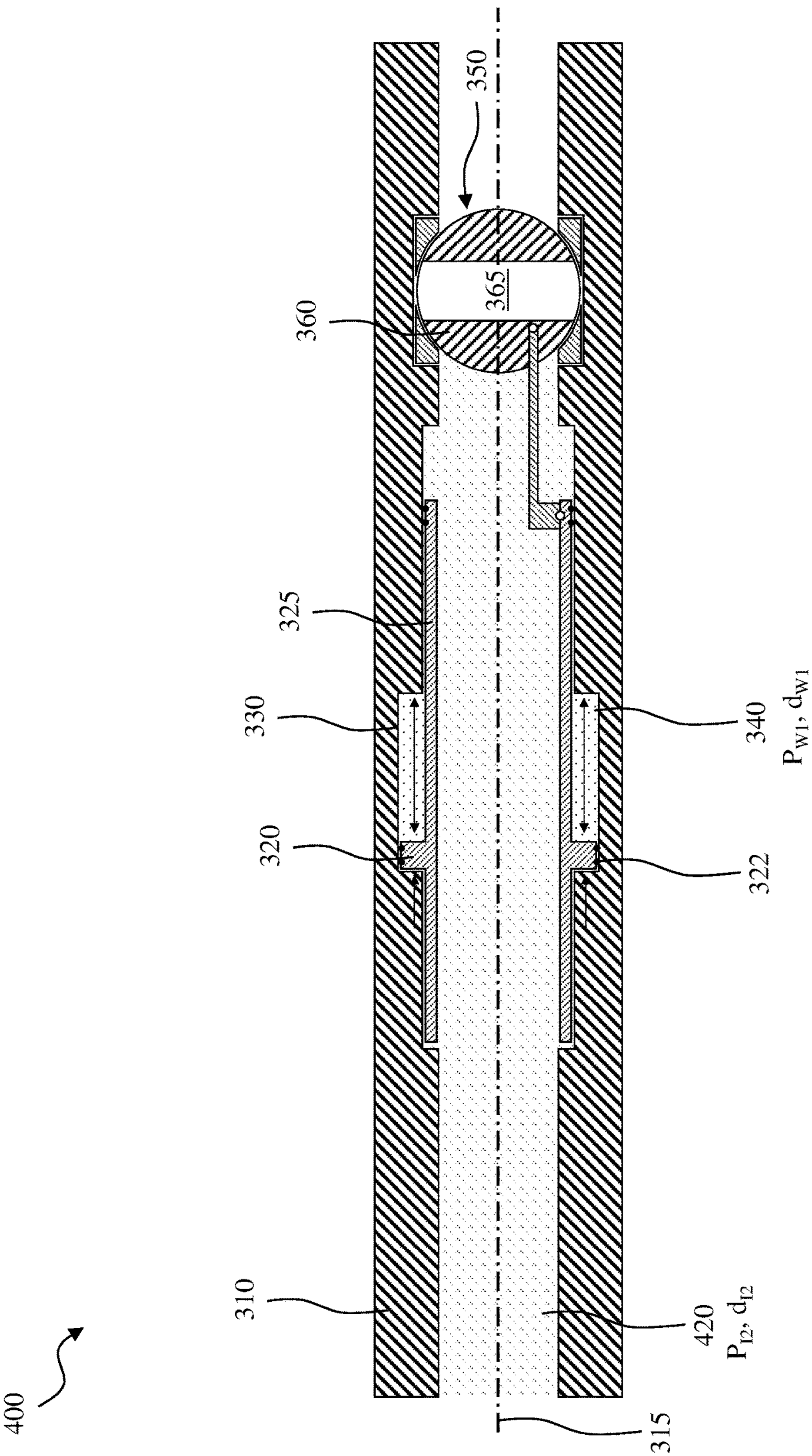


FIG. 4C

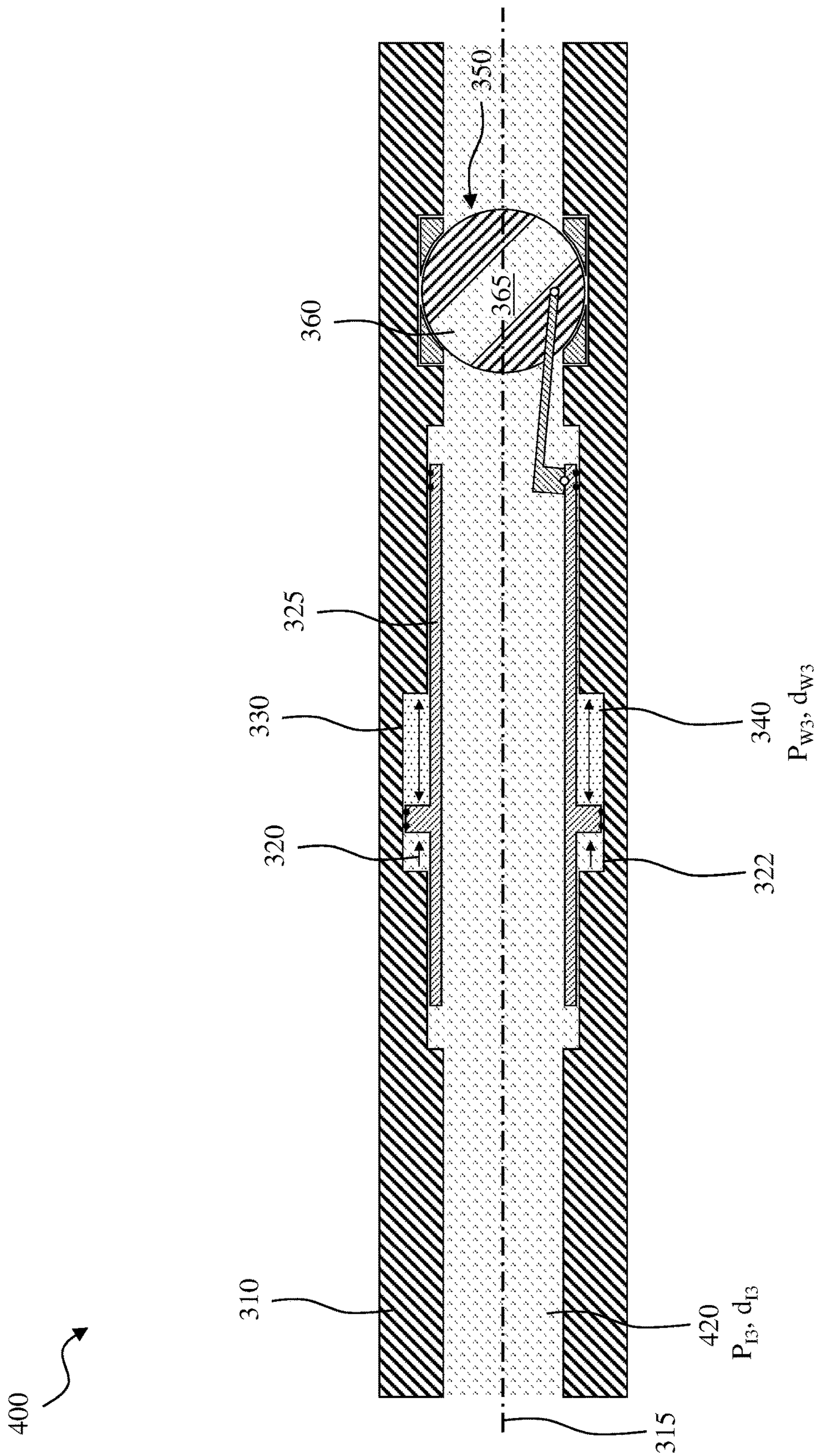


FIG. 4D

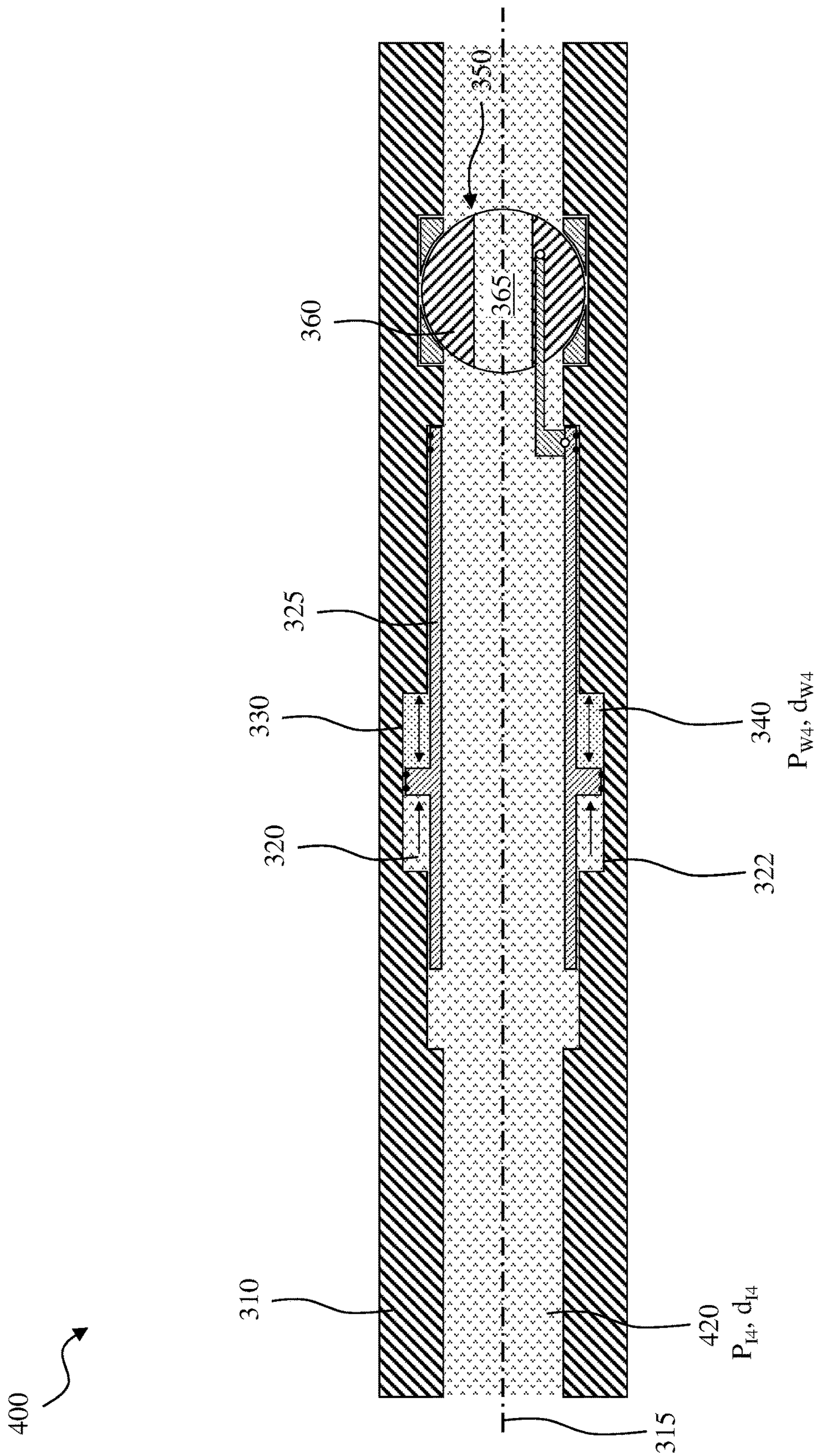


FIG. 4E

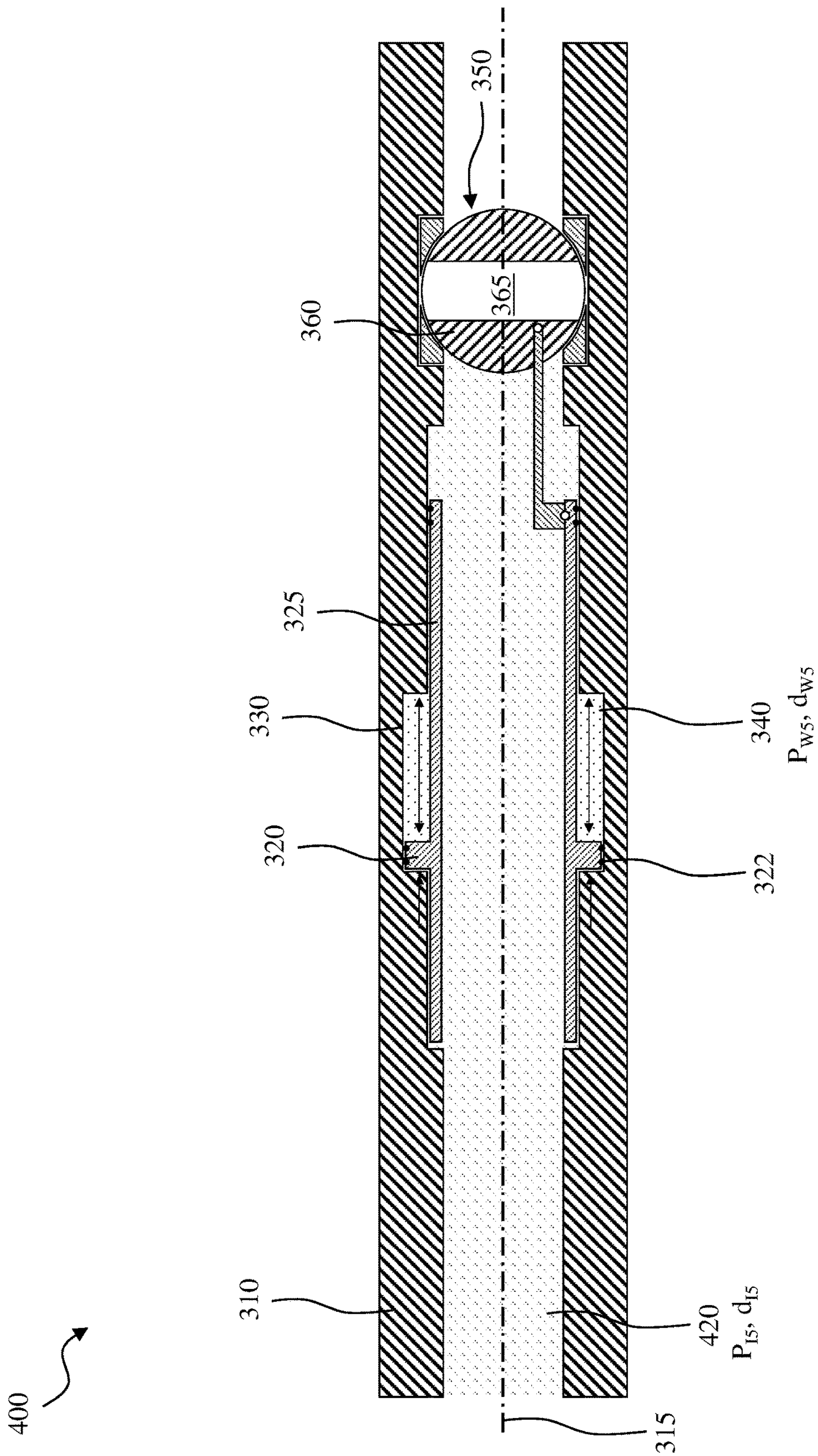
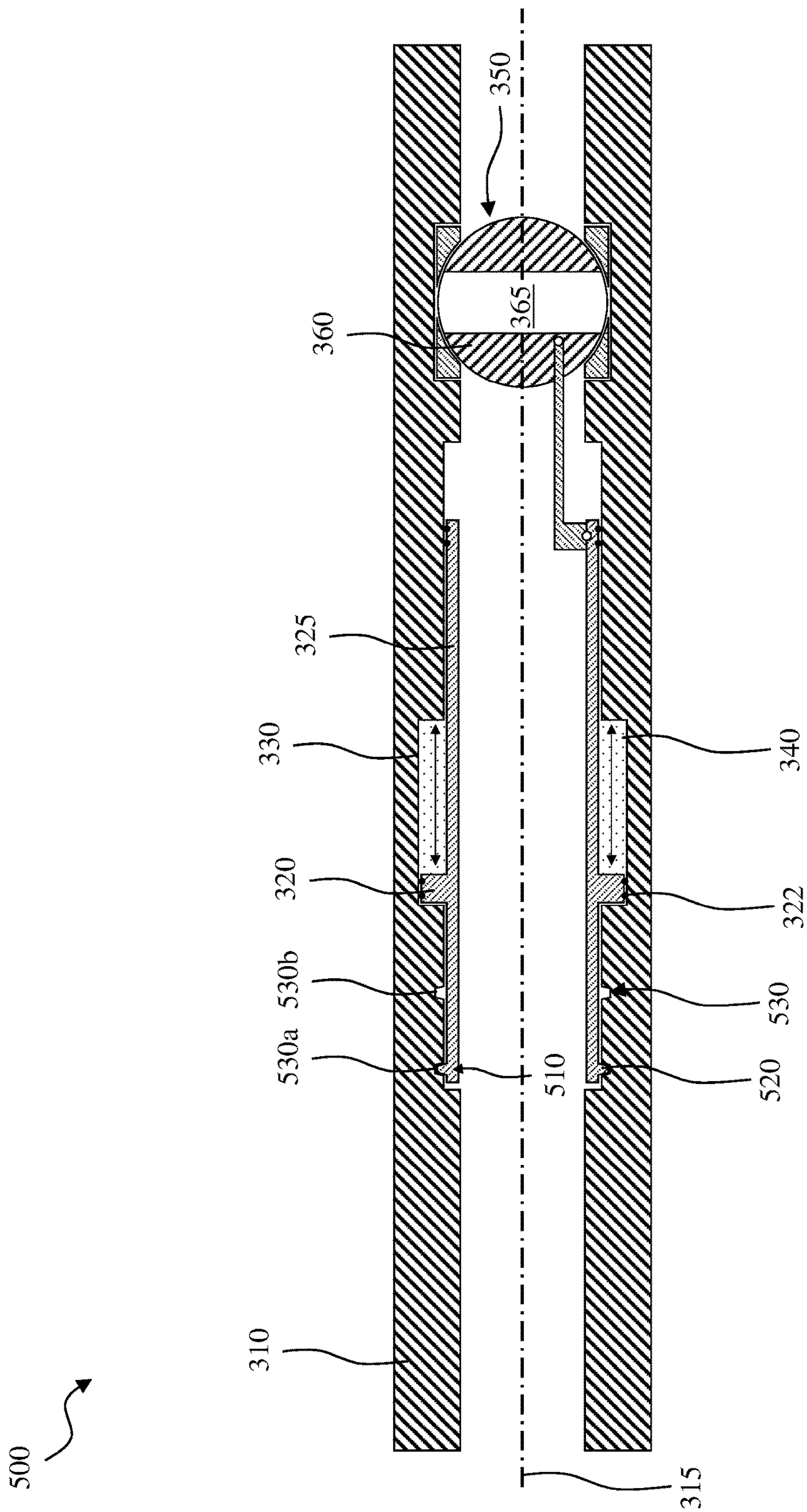


FIG. 4F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365
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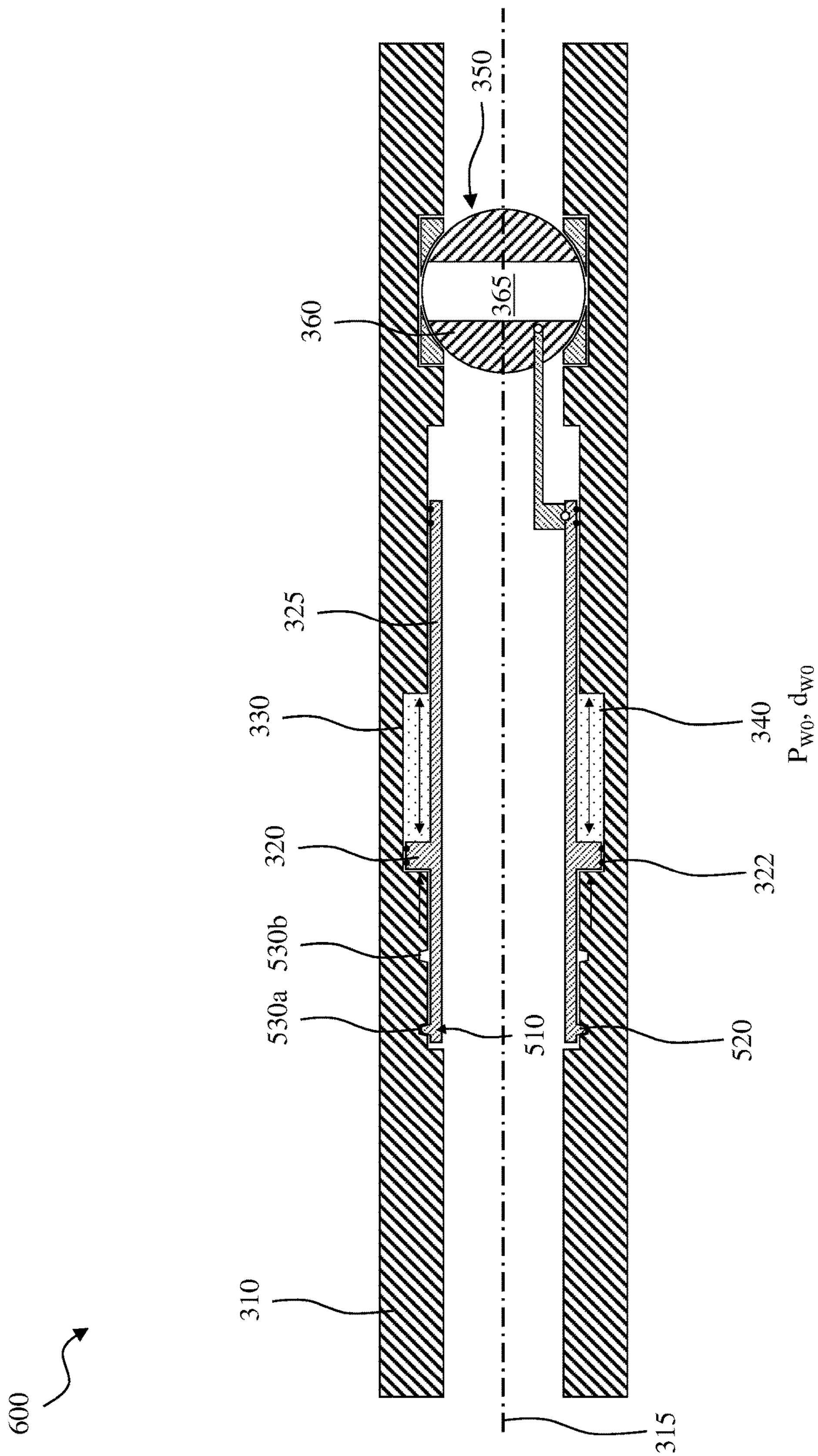
320
322
530b
530
530a
510
520
310
500
315

FIG. 5



350
365
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330
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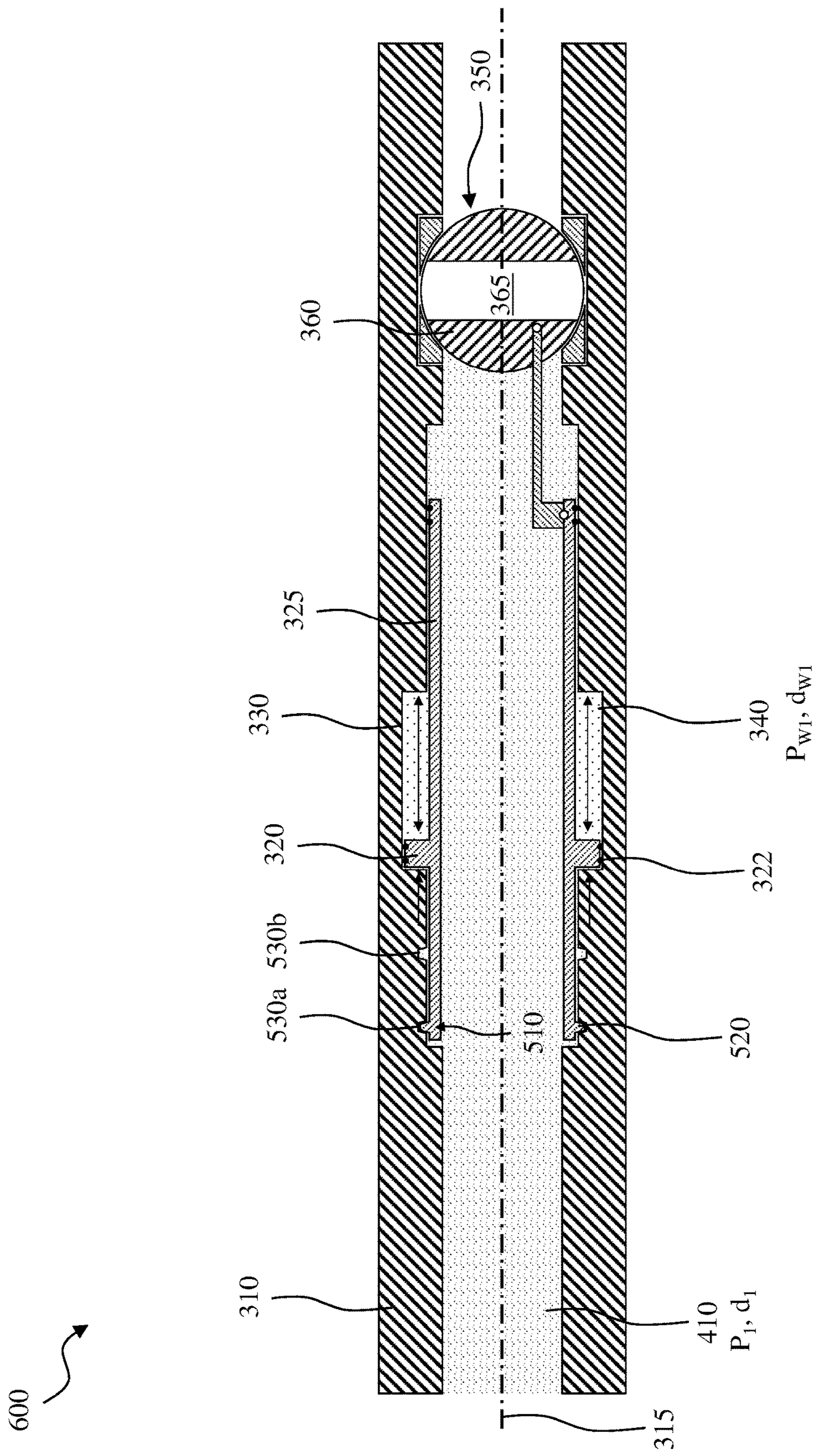
P_{W0}, d_{W0}
320
322
530b
530a
510
520
310
600
315

FIG. 6A


600
310
315
410
P_1, d_1
510
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530a
530b
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P_{W1}, d_{W1}
325
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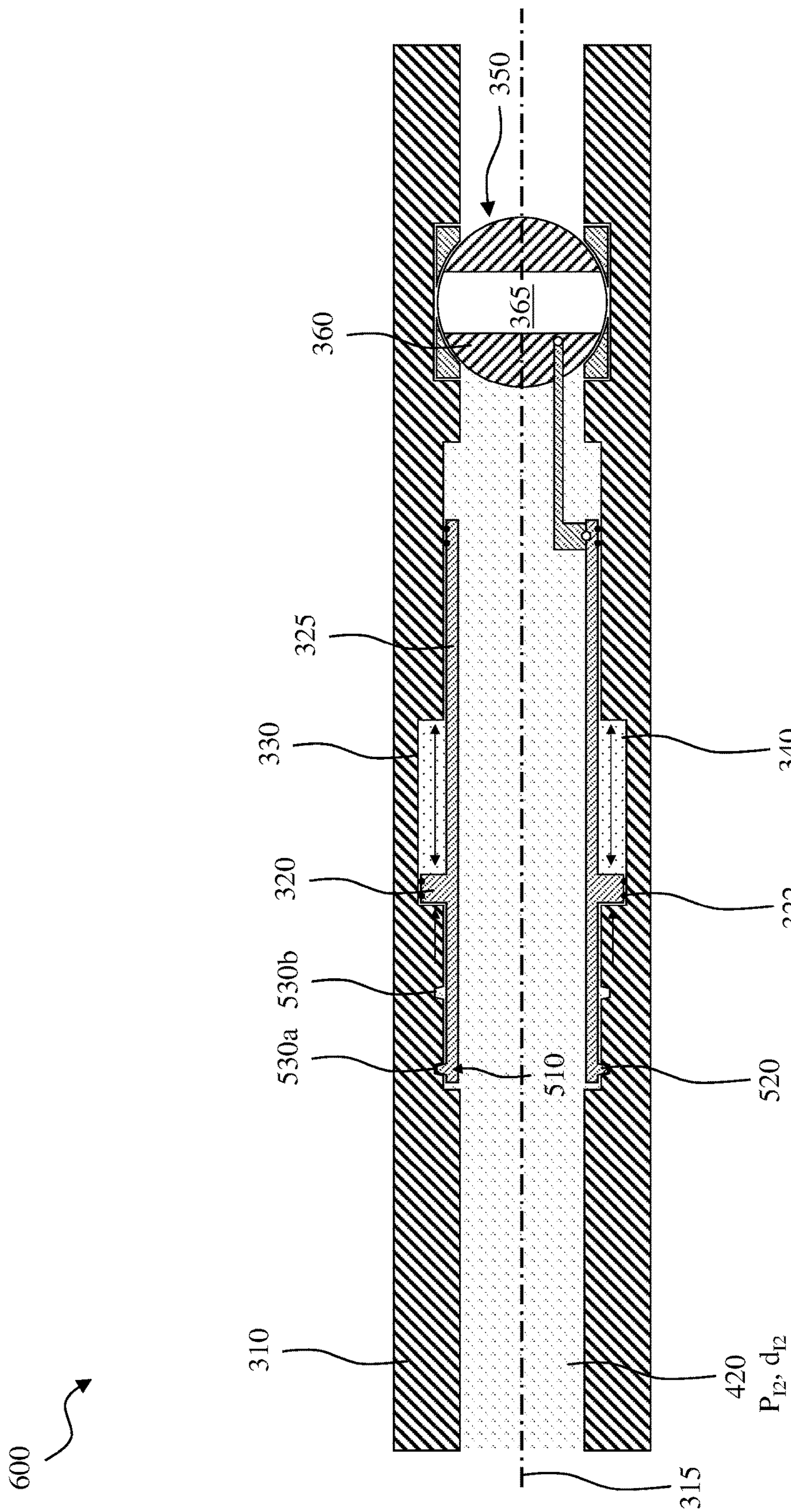
365
350

FIG. 6B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365
360
325
330
340
P_{W1}, d_{W1}
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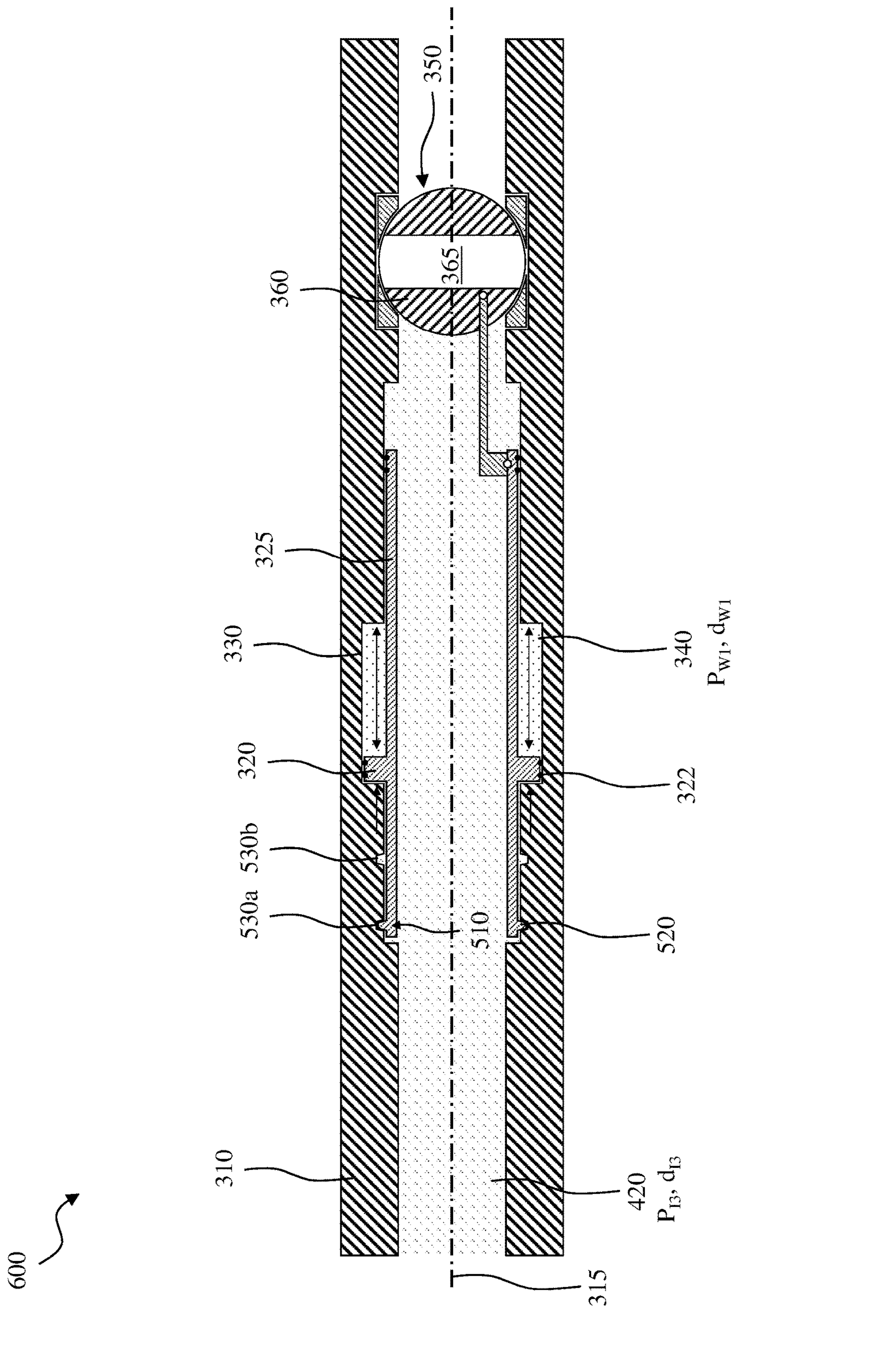
322
530b
530a
510
520
310
420
P_{I2}, d_{I2}
600
315

FIG. 6C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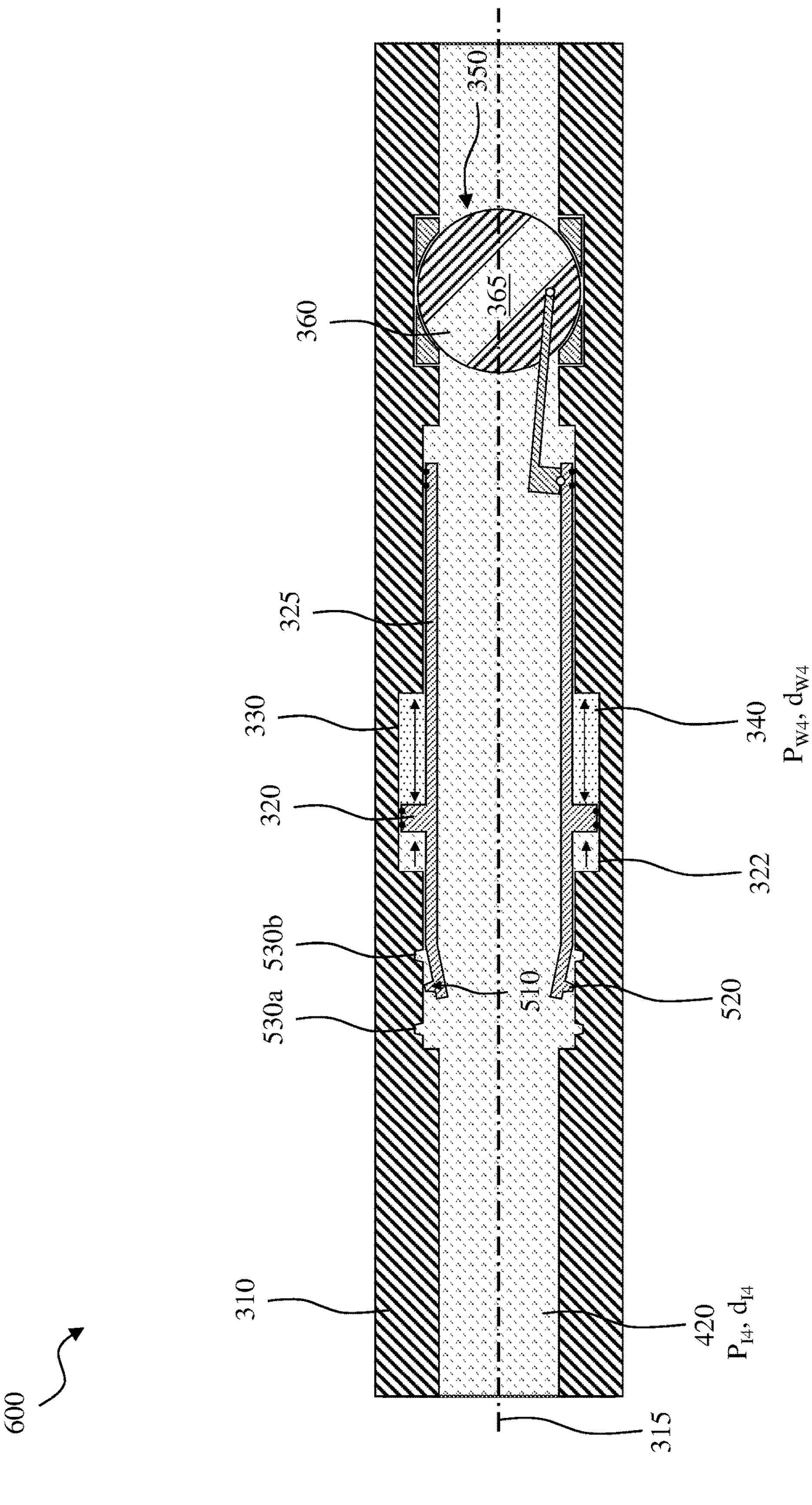
P_{I3}, d_{I3}
315
600

FIG. 6D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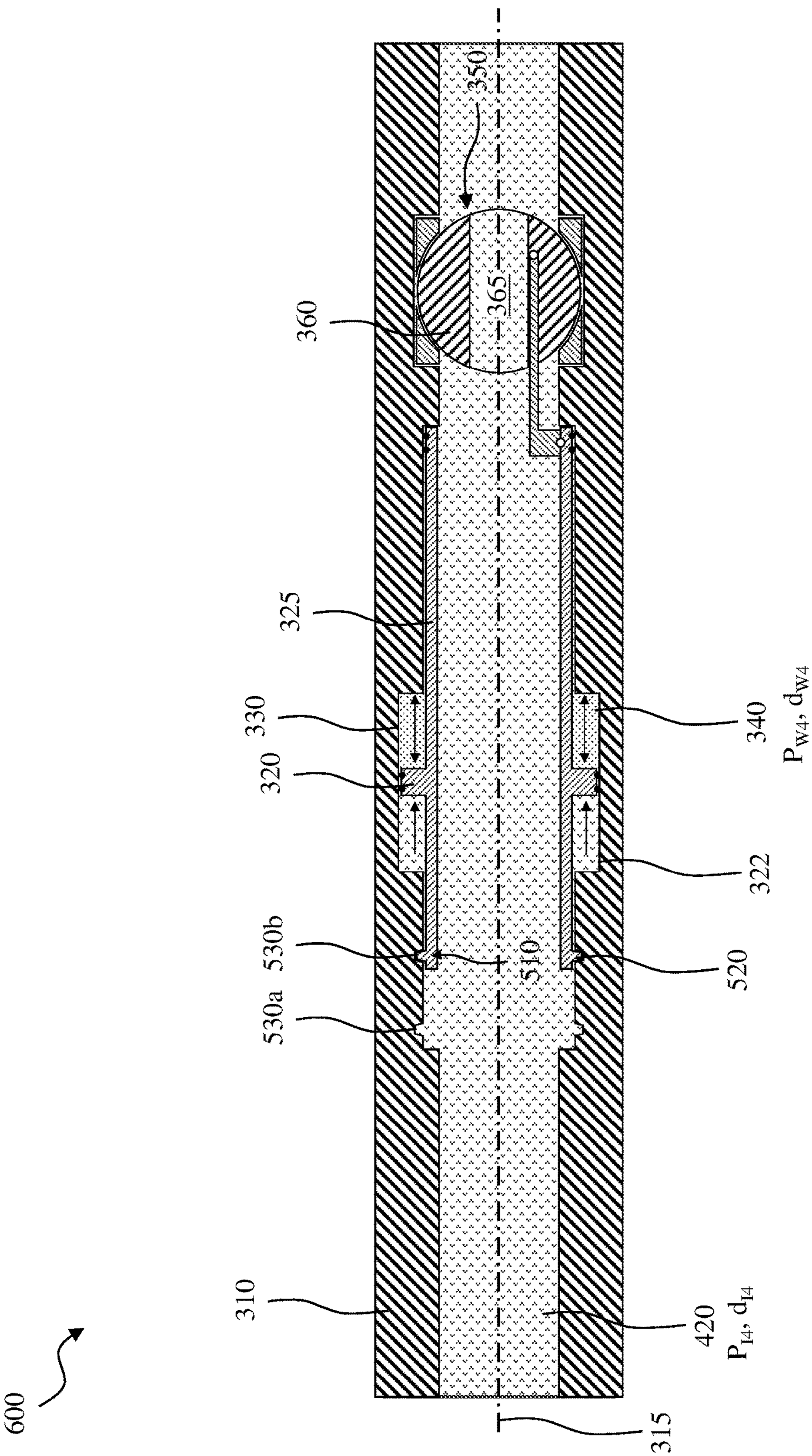
315
600

FIG. 6E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530a
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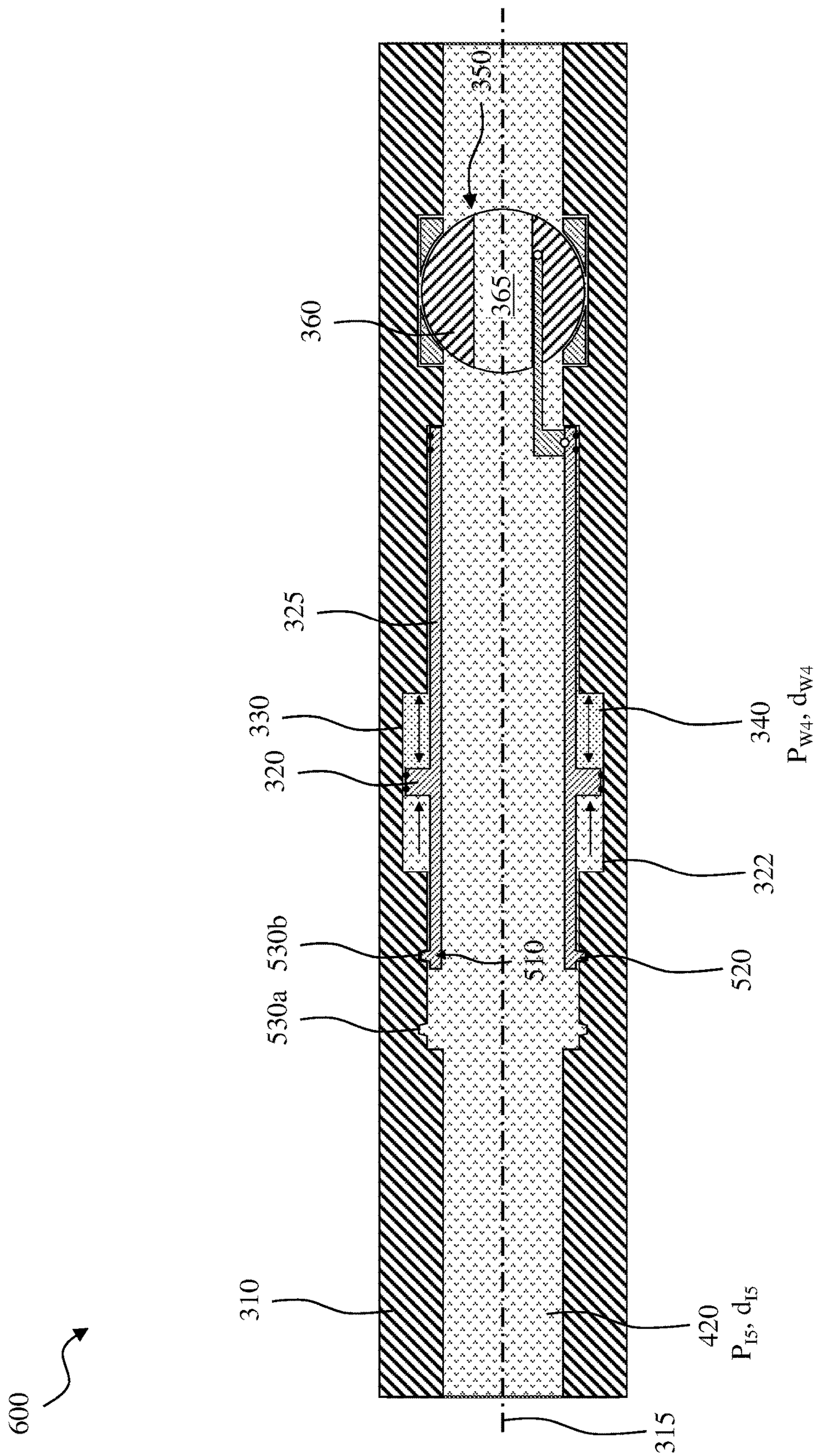
P_{I4}, d_{I4}
315
600

FIG. 6F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P_{I5}, d_{I5}
530a
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P_{W4}, d_{W4}
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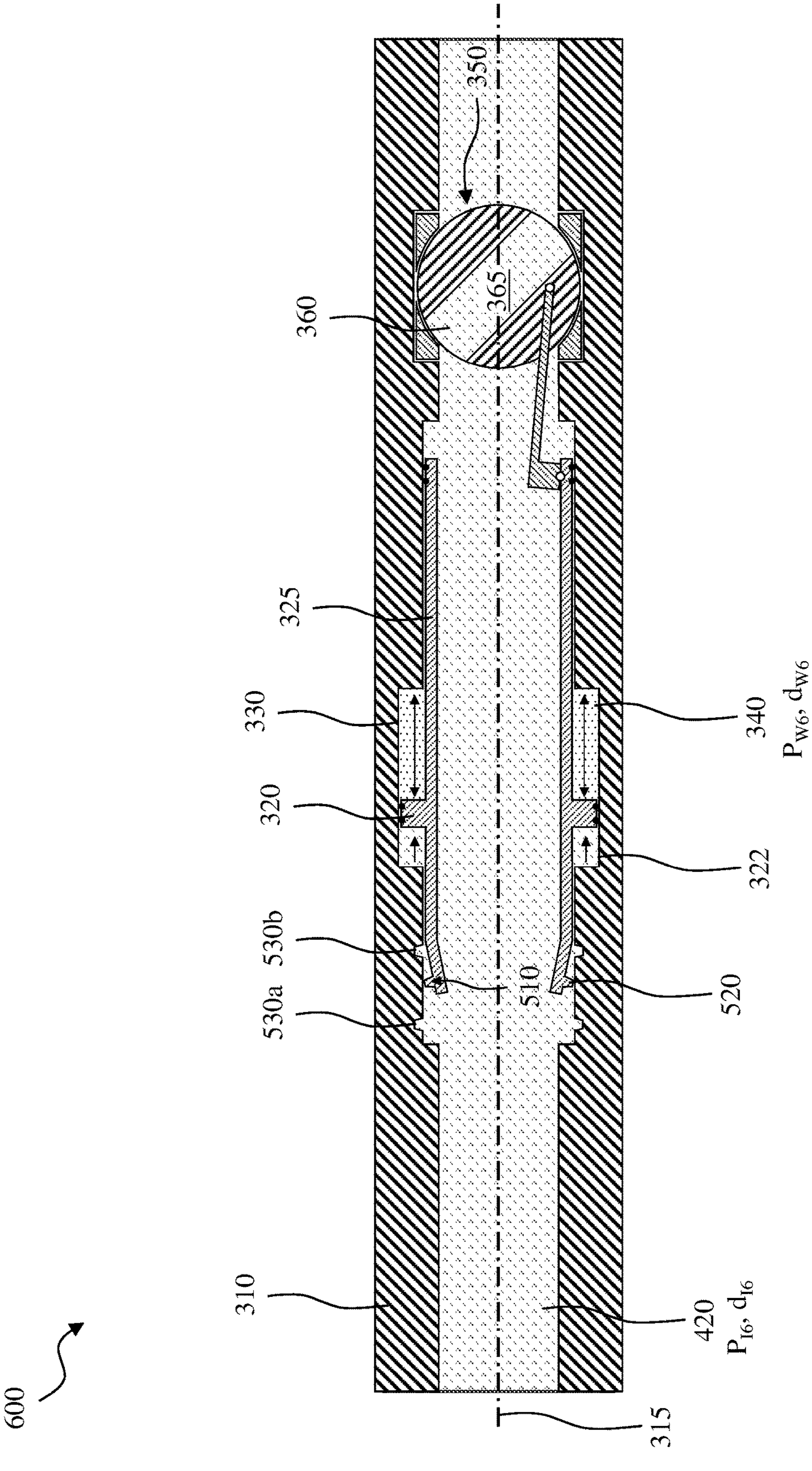
360
365
350

FIG. 6G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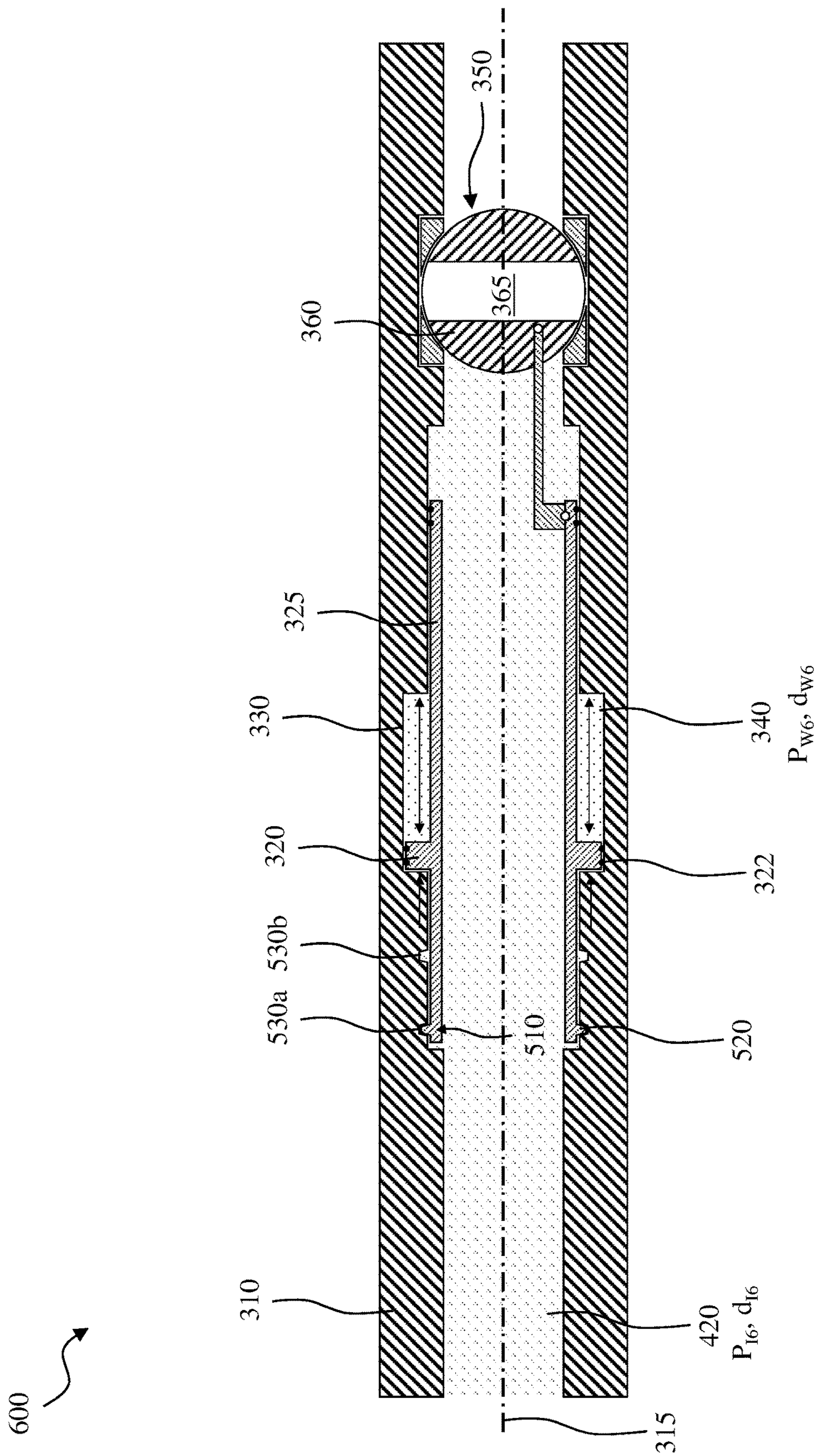
P_{I6}, d_{I6}
600
315

FIG. 6H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P_{W6}, d_{W6}
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322
530b
530a
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310
420
P_{I6}, d_{I6}
315
600

FIG. 6I

AUTONOMOUS INJECTION FLOW CONTROL VALVE

BACKGROUND

[0001] Various subterranean reservoirs, such as wellbores, wellbore networks, and underground spaces such as rock formation material or a subterranean cavity, may be used to store various types of liquids and gases. For example, some subterranean reservoirs may be used for carbon dioxide (CO_2) sequestration, which for example may be utilized to secure a space for permanent disposal of CO_2 based on meeting various environmental/governmental regulations regarding the handling of CO_2. In other examples, a reusable resource, such as hydrogen, may be stored on a temporary basis in a subterranean reservoir, and recovered at a later time for use as needed.

BRIEF DESCRIPTION

[0002] Reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0003] FIG. **1**A illustrates a well system designed, manufactured and/or operated according to one or more embodiments of the disclosure;

[0004] FIG. **1**B illustrates a well system designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure;

[0005] FIG. **2**A illustrates a phase diagram for CO_2;

[0006] FIG. **2**B illustrates a phase diagram for H_2;

[0007] FIG. **3** illustrates a cross-sectional view of a flow control valve designed, manufactured and/or operated according to one or more embodiments of the disclosure;

[0008] FIGS. **4**A through **4**F illustrate different operational states of a flow control valve designed, manufactured and/or operated according to one or more embodiments of the disclosure;

[0009] FIG. **5** illustrates a cross-sectional view of a flow control valve designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure; and

[0010] FIGS. **6**A through **6**I illustrate different operational states of a flow control valve designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure.

DETAILED DESCRIPTION

[0011] In the drawings and descriptions that follow, like parts are typically marked throughout the specification and drawings with the same reference numerals, respectively. The drawn figures are not necessarily, but may be, to scale. Certain features of the disclosure may be shown exaggerated in scale or in somewhat schematic form and some details of certain elements may not be shown in the interest of clarity and conciseness. The present disclosure may be implemented in embodiments of different forms. Specific embodiments are described in detail and are shown in the drawings, with the understanding that the present disclosure is to be considered an exemplification of the principles of the disclosure, and is not intended to limit the disclosure to that illustrated and described herein. It is to be fully recognized that the different teachings of the embodiments discussed herein may be employed separately or in any suitable combination to produce desired results. Moreover, all statements herein reciting principles and aspects of the disclosure, as well as specific examples thereof, are intended to encompass equivalents thereof. Additionally, the term, "or," as used herein, refers to a non-exclusive or, unless otherwise indicated.

[0012] Unless otherwise specified, use of the terms "connect," "engage," "couple," "attach," or any other like term describing an interaction between elements is not meant to limit the interaction to direct interaction between the elements and may also include indirect interaction between the elements described.

[0013] Unless otherwise specified, use of the terms "up," "upper," "upward," "uphole," "upstream," or other like terms shall be construed as generally away from the bottom, terminal end of a well, regardless of the wellbore orientation; likewise, use of the terms "down," "lower," "downward," "downhole," or other like terms shall be construed as generally toward the bottom, terminal end of a well, regardless of the wellbore orientation. Use of any one or more of the foregoing terms shall not be construed as denoting positions along a perfectly vertical or horizontal axis. Unless otherwise specified, use of the term "subterranean formation" shall be construed as encompassing both areas below exposed earth and areas below earth covered by water, such as ocean or fresh water.

[0014] The description that follows includes example systems, methods, techniques, and program flows that embody aspects of the disclosure. However, it is understood that this disclosure may be practiced without these specific details. For instance, this disclosure refers to use of carbon dioxide (CO_2) and hydrogen (H_2) in illustrative examples. Aspects of this disclosure can also use other types of gases, including combinations of one of those gases with another gas. In other instances, well-known instruction instances, protocols, structures and techniques have not been shown in detail in order not to obfuscate the description.

[0015] Example embodiments can be used as part of CO_2 sequestration in the subsurface formation. In some embodiments, a flow control valve is positioned to restrict injection of CO_2 when the CO_2 is in a low-density phase state, (e.g., a gas), and minimally restrict injection or fully allow injection when the CO_2 is in a high-density phase state, (e.g., a liquid). Other example embodiments may be used for storage of hydrogen (H_2) that can be an energy storage option, in some embodiments where H_2 can be injected and produced from a same wellbore.

[0016] Some embodiments can include a vapor-transition flow control valve that allows injecting of CO_2 in some embodiments, or hydrogen H_2 in other embodiments, into a formation or cavity. The vapor-transition flow control valve can ensure that the injected fluid is in a desired phase state, such as a liquid or a high density fluid, by allowing a flow of the fluid to be injected through or past the flow control valve only when the fluid that is provided to the valve is in the desired phase state, while blocking the flow of the fluid through the flow control valve when the fluid to be injected into the formation or cavity is not provided to the valve in the desired phase state. This can be important to ensure that the flow velocities of the injected fluid are low, the temperature of the injection fluid is appropriate, that control of the reservoir is maintained, and that a consistent flow distribution of the injected fluid is achieved in the wellbore.

[0017] Example embodiments can include an axially sliding piston-operated valve driven by a phase changing of a fluid contained within a sealed working fluid chamber of the flow control valve, (referred to as a working fluid or as a fill fluid). The axially sliding piston-operated valve is configured to be deployed in a borehole formed in a subterranean formation. For example, the axially sliding piston-operated valve can be integrated with production tubing that is positioned in the borehole as part of downhole operations. When the injection pressure of the fluid to be injected into the formation or cavity is high, the working fluid included in the sealed working fluid chamber can be compressed via the axially sliding piston and transition into a high density fluid phase state, thereby axially sliding a control arm to actuate a valve assembly of the flow control valve to an "open" configuration. With the valve assembly of the flow control valve in the "open" configuration, injection fluid provided to the flow control valve is allowed to pass the flow control valve and be injected into the subterranean formation. When the injection pressure of the fluid to be injected into the formation or cavity is low, the working fluid included in the valve can be less compressed, and transition into a low density fluid phase state, thereby axially sliding the control arm to actuate the valve assembly of the flow control valve to a "closed" configuration. With the valve assembly of the flow control valve in the "closed" configuration, injection fluid provided to the flow control valve is blocked from passing through or around the flow control valve, and thereby is blocked from being injected into the subterranean formation. Likewise, fluid already contained within the subterranean formation is prevented from flowing into the tubing.

[0018] In some embodiments, the working fluid used in the axially sliding piston-operated valve is selected to have a same (or a substantially similar) vaporization curve as compared to the fluid to be injected into the formation or cavity where the flow control valve is being employed. In some embodiments, the working fluid and the fluid to be injected can be defined as "substantially similar" if a variation in their vaporization curves is within a threshold. For example, the working fluid or fill fluid and the fluid to be injected can be defined as substantially similar if a variation in their vaporization curves is less than 1%, 5%, 10%, 25%, etc. In some embodiments, the working fluid and the fluid to be injected can be defined as substantially similar if the difference in their vapor transition is less than a threshold. For example, the working fluid and the fluid to be injected can be defined as substantially similar if the difference in their vapor transition is within 100 pounds per square inch (psi), 50 psi, 200 psi, etc.

[0019] In some embodiments, the working fluid and the fluid to be injected can be defined as the same or substantially similar based on their chemical composition. For example, the working fluid to be injected can have the same chemical composition, for example, both are CO_2, or both are H_2. In some embodiments, the working fluid and the fluid to be injected can be defined as substantially similar if the difference in their chemical composition is less than a threshold. For example, the working fluid and the fluid to be injected can be defined as substantially similar if the difference in their chemical composition is less than 1%, 5%, 10%, 25%, etc.

[0020] In some embodiments, the working fluid can be an azeotrope fluid. The azeotrope fluid in the sealed working fluid chamber of the axially sliding piston-operated valve may be a combination of fluids such that the transition temperature of the combined azeotrope fluids will be at a slightly lower temperature or a slightly higher temperature in relation to the injection fluid, and/or having a slightly lower pressure or slightly higher pressure than the injection fluid that is to be controlled using the flow control valve. In some embodiments, the working fluid can be at the same pressure as the fluid to be injected into the formation. In such embodiments, the phase change in the working fluid occurs at the same time as the injected gas within the tubing.

[0021] In some embodiments, the term "liquified" and "gaseous" can be used to describe the different phases of gas. Above the critical point, the gas is considered to be a supercritical fluid. Thus, the volume change can be used to create a valve that closes if the injection pressure is insufficient to inject a high-density fluid. As used in this disclosure, a "high density fluid" is any fluid at a temperature and pressure above its critical point or in its liquid form, and a "low density fluid" is any fluid in its vapor or gaseous phase with temperature and pressure below its critical point.

[0022] Also, while described in reference to being positioned in a production tubing for fluid injection into the subsurface formation (e.g., one that closes on the tubing string, such as a main tubing string, rather than on a sidewall), in some other embodiments, the valve can be used in other downhole configurations and applications. For example, the valve can be positioned in the well using traditional intervention techniques for fluid injection into the subsurface formation. In another example, such valves can be used axially along the borehole where they are installed using bridge plugs or lock mandrels to manage different zones into which fluid can flow.

[0023] Additionally, in some embodiments, a system can include multiple flow control valves positioned at different locations (e.g., different depth locations within the wellbore). In some embodiments, one or more of these flow control valves can have independent flow paths for injection into the formation. In some embodiments, one or more of these flow control valves may have different vaporization transitions. In some embodiments, multiple flow control valves can be installed into a single device at a given location along the wellbore. The flow control valves can be configured differently so that the number of open and closed flow control valves can be a function of downhole conditions at the device, which in turn is a function of surface injection conditions. Such embodiments can regulate flow when surface injection conditions are varied. Similarly, in some embodiments, multiple flow control valves can be positioned axially along the wellbore with a same or different configuration.

[0024] While described in reference to a phase valve, example embodiments can use any type of flow restriction in order to ensure that the injection fluid is in a high density fluid phase. Examples of flow restrictors that can be used include an inflow control device (ICD) (such as a nozzle, venturi, porous media, or tube), an autonomous inflow control device (AICD), autonomous inflow control device (AICV), a wireless smart well node, etc. In some embodiments, the injected fluid is achieving critical flow by moving at a sonic velocity with the flow restriction.

[0025] Furthermore, a device according to the disclosure can be installed in an already existing tubing string. In at least one embodiment, a device according to the disclosure could be installed on a lock nipple or a profile within the wellbore, among other locations. Additionally, in at least one other embodiment, a device according to the disclosure may be placed in the upper completion, while injection components are located in the lower completion.

[0026] FIG. **1**A illustrates a diagram of a well system **100** configured for fluid injection into a subterranean formation, according to various embodiments of the disclosure. Although described below with respect to a well system configured to perform fluid injection of CO_2 into a formation, embodiments of the well system **100** are not limited to operations involving CO_2 injection, and may include fluid injection operations including other types of gases having various chemical compositions. As shown in FIG. **1**A, various components including a storage reservoir or vessel **104**, a fluid pump **105**, and a wellhead **110** are located above a surface **101**, and proximate a wellbore **102** extending below surface **101** into a subterranean formation **108**.

[0027] Vessel **104** may be any type of vessel configured to contain the CO_2 that is to be injected into formation **108** using the well system **100** for permanent storage of the CO_2 in the formation. The use of the phrase "permanent storge" is not necessarily in reference to a particular timeframe, but refers to storage of the CO_2 without the intent to retrieve the CO_2 from the formation at some time in the future. The CO_2 contained in vessel **104** may have been produced from an oil and gas reservoir, generated for example as a result of a manufacturing process, or from some other manmade source, and was captured and placed in vessel **104** as an alternative to letting the CO_2 be emitted into the atmosphere. System **100** is configured to take the CO_2 contained in vessel **104**, and to inject the CO_2 into formation **108** for permanent storage within the formation, as further described below.

[0028] As illustrated in FIG. **1**A, vessel **104** is coupled to fluid pump **105** through fluid conduit **103**, wherein fluid pump **105** is coupled to wellhead **110** through fluid conduit **109**. In various embodiments, fluid pump **105** includes pump **107** configured to pump CO_2 provided through fluid conduit **103** to the wellhead **110** through fluid conduit **109**. In various embodiments, pump **107** is powered by a pump driver, such as motor **106**, which in various embodiments is an electric motor. The CO_2 provided to wellhead **110** is coupled through valve inlet **112** to one or more valves **111** configured to controllably couple a flow of fluid from the valve inlet **112** to the valve outlet **113**. A controller **115**, which may comprise a computing device with one or more processors and other computing devices, such a computer memory, may be coupled to devices such as the one or more valves **111**, and is configured to control the operation of the one or more valves **111**. In various embodiments, the controller **115** may also be configured to control the operation of the fluid pump **105** in order to regulate the pressure and/or the flow rate of CO_2 being provided to the wellhead **110** from vessel **104**.

[0029] In various embodiments, fluid pump **105** may further include temperature control devices **114**, which may include heating elements and/or a chiller/compressor unit configured to heat or cool, respectively, the fluid being provided to the wellhead **110**, via heating and/or cooling. Heating or cooling of the fluid, in conjunction with the use of pump **105** to pressurize the fluid, may be controlled, in some embodiments by controller **115**, in order to place the fluid into a desired and predetermined phase state for injection into the formation **108**.

[0030] In well system **100**, valve outlet **113** is coupled to be in fluid communication with a downhole fluid tubing (e.g., tubing) **121**, which extends down into wellbore **102** and is enclosed within a tubing string **120**. Tubing string **120** includes a hollow center passageway through which downhole fluid tubing **121** extends. Tubing string **120** is also physically coupled to one or more packers and to one or more flow control valve assemblies, which tubing string **120** helps secure within the wellbore **102**. As shown for the well system **100**, tubing string **120** extends from surface **101**, and is positioned within and is encircled by upper casing **122**, which also extends from surface **101** to some depth within wellbore **102** along a longitudinal axis **139** of the borehole. In various embodiments, at least some portion of the upper casing **122** may be encased in cement **123**. In addition, one or more centralizers **119** may be positioned within the upper casing **122**, the centralizers **119** configured to extend between the inner surface of the upper casing **122** and an outer surface of the tubing string **120**, and thus stabilize the tubing string **120** with the upper casing.

[0031] In various embodiments, a packer **124** is positioned within upper casing **122** at some predetermined depth within wellbore **102**, the packer **124** coupled to tubing string **120** and encircled by a sealing element **125** the extends between the packer **124** and the inner surface of the portion of the upper casing **122** where the packer **124** is positioned. A first flow control valve **140** is positioned downhole from packer **124**, and is physically coupled to tubing string **120**. The first flow control valve **140** includes one or more axially sliding pistons, as described herein, configured to control a flow of CO_2 through the first flow control valve **140** and into formation **108** in the areas of the formation **108** proximate to the location of first flow control valve **140** within the wellbore **102**. Packer **124**, in conjunction with sealing element **125**, provide isolation of an annulus **132**, which surrounds the flow control valve **140**, from annulus **130**, which extends from surface **101** to the uphole side of packer **124** and encircles the tubing string **120** within upper casing **122**.

[0032] Internal fluid passageways within the first flow control valve **140** are in fluid communication with downhole fluid tubing **121**, wherein the one or more axially sliding pistons are configured to controllably couple the internal fluid passageways of first flow control valve **140** to one or more passageways of the first flow control valve **140**. As such, the one or more axially sliding pistons of the first flow control valve **140** are configured to allow CO_2 that is received at the first flow control valve **140** from the downhole tubing string **121** to be controllably released through the passageways and into annulus **132**. The pressure of the CO_2 released into annulus **132** drives the CO_2 through perforations **147** extending along wellbore **102** in the vicinity of the first flow control valve **140**, and out into formation **108**, as illustratively represented by arrows **133**. In various embodiments, only one flow control valve, such as first flow control valve **140**, is included in well system **100**, wherein the CO_2 released from the passageways is configured to fill the borehole extending below packer **124**, and to exit the borehole through one or more sets of perforations **147** for injection into formation **108**.

[0033] In the alternative, in the well system **100** as illustrated in FIG. **1**A, three sets of flow control valves, **140**, **142**, and **144** are shown, the flow control valves spaced apart from one another along the wellbore **102**, and wherein each of the flow control valves is surrounded by a respective annulus (**132**, **134**, **136**), which are isolated from one another by packers **126** and **128**, and wherein the upper-most annulus **132** surrounding the first flow control valve **140** is isolated from the annulus **130** extending to surface **101** by packer **124**. Each of the flow control valves are in fluid communication with tubing **121**, and thus are configured to receive a flow of fluid being provided from the surface **101** through tubing **121**. As illustrated in FIG. **1**A, the passageway of the first flow control valve **140** may be directed to perforations **147** adjacent to annulus **132**, and further directed into formation **108** in a zone generally indicated as zone **150**. Similarly, the passageways of the second flow control valve **142** may be directed to perforations **147** adjacent to annulus **134**, and further directed into formation **108** in a zone generally indicated as zone **151**, while the passageways of the third flow control valve **144** may be directed to perforations **147** adjacent to annulus **136**, and further directed into formation **108** in a zone generally indicated as zone **153**. As such, the arrangement of the flow control valves and the packers isolating the annuluses surrounding the flow control valves, respectively, may contribute to better control and dispersion of the fluid into formation **108** along the entirety of the wellbore **102** designated for fluid injection.

[0034] The arrangement of the flow control valves and packers as shown in FIG. **1**A is one non-limiting example of a well system, such as well system **100**, and other variations are possible and are contemplated for use in various embodiments of well system **100**. For example, the number of flow control valves included in a particular well system is not limited to a particular number of flow control valves, and may include one or more flow control valves. Examples of well system **100** are not limited to having the flow control valves positioned at a particular depth from surface **101**, and may include flow control valves positioned at varying depths, for example based on the location of a formation material that is determined to be useful for the storage of a fluid, such as CO_2, within the formation material. In various embodiments, the flow control valves that are included in a well system **100** may or may not be evenly spaced relative to one another along the wellbore, and may include groups of one or more flow control valves that are spaced apart from another group of flow control valves by a distance along the wellbore that is different from the spacing between other groups of valve assemblies.

[0035] In various embodiments of a well system, a single packer may be used to isolate the annuluses surrounding each of the flow control valves included in the well system from the annulus extending to the surface of the well system. In various embodiments, each of the flow control valves included in a well system may be isolated from the other flow control valves included in the well system by a set of packers positioned uphole and downhole from the location of each of the flow control valves. In various embodiments, a group of two or more flow control valves may be isolated by a pair of packers so that the group of two or more flow control valves is configured to be in fluid communication with a common annulus. These and other variations of the flow control valves and packer arrangements are possible and are contemplated for use in configurations of well systems that may be utilized for fluid injection operations as described herein, and any equivalents thereof. Further, while wellbore **102** is shown as comprising a vertically oriented borehole, embodiments of wellbores where the valve assemblies may be deployed are not limited to wellbores having any particular orientation, and may include vertical, horizontal, and/or inclined wellbores, and combination of these, including well systems including one or more branches coupled to a main, a secondary, or other network(s) of a wellbore.

[0036] In operation, CO_2 stored in vessel **104** is pumped to the wellhead **110** by fluid pump **105** in a phase state that is desirable for injection of the CO_2 into formation **108**. In various embodiments, that desired state includes CO_2 in a high-density fluid phase. In various embodiments, the temperature of the fluid received from the vessel **104** may be altered by one or more temperature control devices **114** in order to allow the pump **107** to pressurize the fluid while allowing the fluid to be transformed into and/or maintained in a desired phase state for injection into formation **108**. The high-density fluid CO_2 is coupled through valves **111** and into the downhole fluid tubing **121**, where it passed through the downhole fluid tubing **121** and is provided to each of the flow control valves **140**, **142**, **144**.

[0037] When the CO_2 is provided to the flow control valves in the desired phase state, the axially sliding piston included in the flow control valves is/are configured to actuate to an "open" configuration, allowing the CO_2 to flow through or past the valve assembly, and exit through one or more of fluid passageways, to a respective annulus, and then to flow through perforations **147** in the casing liner of the wellbore and into the formation **108**. Arrows **133** represent the flow of CO_2 exiting the first flow control valves **140** and flowing into formation **108** in zone **150**. Arrows **135** represent the flow of CO_2 exiting passageways of the second flow control valve **142**, and flowing into formation **108** in zone **151**. Arrows **137** represent the flow of CO_2 exiting passageways of the third flow control valve **144** and flowing into formation **108** in zone **153**.

[0038] In the event the CO_2 arriving at the flow control valves **140**, **142**, and **144** via the downhole fluid tubing **121** is not in the desired phase state, for example is in a low density phase, the axially sliding pistons are configured to actuate to a "closed" configuration, and to block the flow of the CO_2 from passing through the flow control valves and into the formation **108**. In addition, if the fluid pressure present in the formation rises to a level that exceeds the pressure present in the CO_2 arriving at the valve assembles from the surface, the phase of the CO_2 present in the backflow will again cause the axially sliding pistons to actuate to the "closed" position, thereby preventing the CO_2 from escaping from the formation **108** back through the flow control valves, and in conjunction with the packers present in the wellbore, from escaping from the formation **108** back through the wellbore. Thus, the axially sliding pistons of the flow control valves are configured to allow a flow of CO_2 provided to the flow control valves in the desired phase state to be distributed and injected into formation **108**, while blocking the flow of CO_2 into the formation when the CO_2 provided to the flow control valves from the surface is not in the desired phase state.

[0039] In various embodiments, the tubing string **120** is configured to be removable, along with the flow control valves and/or the packers coupled to the tubing string, upon completion of the fluid injection operations that are to be performed on wellbore **102**. In various embodiments, after removal of tubing string **121**, the wellhead may be sealed off to provide a fluid seal between the wellbore **102** and areas above the surface.

[0040] Turning to FIG. **1**B, illustrated is a diagram of a well system **170** configured for fluid injection into a subterranean formation comprising a cavity, according to various embodiments. Although described below with respect to a well system configured to perform fluid injection of H_2 into a cavity **175**, embodiments of the well system (e.g., system) **170** are not limited to operations involving H_2, and may include fluid injection operations including other types of fluids having various chemical compositions. As shown in FIG. **1**B, the well system **170** includes components above surface **101** and proximate to wellbore **102** that are the same as or similar to components described above with respect to vessel **104**, fluid pump **105**, and wellhead **110**. In various embodiments, these above-surface components are configured to deliver a flow of fluid comprising H_2 through tubing **121** to wellbore **102**. Tubing **121** is coupled to and is in fluid communication with a flow control valve **172**, which includes one or more axially sliding pistons configured to controllably provide a flow of the H_2 provided to the flow control valve **172** to passageway **173**. Any H_2 exiting passageway **173** of the flow control valve **172** is dispensed into the annulus surrounding the flow control valve **172**, and may then be dispersed out the end of casing **122** and into cavity **175**, as illustratively represented by arrows **174**. A packer **124**, in conjunction with sealing element **125**, provides a fluid seal positioned uphole of the flow control valve **172**, and is configured to prevent any H_2 that exits passageway **173** from traveling back uphole past the flow control valve **172** and into annulus **130**.

[0041] The flow control valve **172** may include one or more axially sliding pistons configured with a chamber comprising a working fluid that allows each of the one or more axially sliding pistons to allow a flow of H_2 through the respective flow control valve when the H_2 provided by the tubing **121** to the valve assembly is in the desired phase state, and to block a flow of H_2 through the respective flow control valve when the H_2 provided by the tubing **121** to the valve assembly is not in the desired phase state. In various embodiments, the desired phase state for the H_2 provided by the tubing **121** is a liquid or supercritical phase state (high density fluid), which when provided to the flow control valve is configured to cause the one or more axially sliding pistons of the flow control valve **172** to actuate to an "open" configuration and provide a fluid passageway of the H_2 to flow from tubing **121**, through or around the flow control valves, and to be expelled from the flow control valve and into cavity **175**.

[0042] In various embodiments, the flow control valve **172** is configured so that a minimum pressure is maintained within cavity **175**. In some embodiments, the minimum pressure that is to be maintained in the cavity **175** is in a range of 500 to 5000 pounds/square inch (PSI). In various embodiments, the minimum pressure level to be maintained within cavity **175** is set in order to assure the structural integrity of cavity **175**, and for example to assure against a collapse of any portion of the cavity.

[0043] FIG. **2**A illustrates a phase diagram **200** for CO_2. Phase diagram **200** includes a vertical axis **201** representing temperatures (temperature increases moving upward in the diagram), and horizontal axis **202** representing pressure (increasing pressure moving in the right-hand direction in the diagram). Graphical line **204** represents the vaporization curve for CO_2. For temperature/pressure combinations that fall above graphical line **204**, as illustratively represented by the area **203** within phase diagram **200**, CO_2 is in a low-density gaseous phase state. For temperature/pressure combinations that fall below and to the right of graphical line **204**, as illustratively represented by area **205** within the phase diagram **200**, CO_2 is in a high-density fluid phase state.

[0044] As shown by phase diagram **200**, the vaporization curve represented by graphical line **204** for CO_2 requires increasing temperature as the pressure increases for the CO_2 to be in the low-density fluid phase state. By way of illustrations, phase diagram **200** includes three illustrative points: Point A, Point B, and Point C. At point A, the temperature/pressure combination is below the vaporization curve represented by graphical line **204**, and therefore CO_2 at this temperature/pressure combination is a high density fluid. If the pressure of the CO_2 is reduced to now include a temperature/pressure combination for the CO_2 represented by point A*, point A* is still below the vaporization curve, and therefore CO_2 at the temperature/pressure combination represented by point A* is still a high density fluid, and has not changed phase state relative to CO_2 at point A due to the lowering of the temperature represented by the change in the temperature/pressure combination from point A to point A*.

[0045] At point B, the temperature/pressure combination is below the vaporization curve represented by graphical line **204**, and therefore CO_2 at this temperature/pressure combination is a high density fluid. If the pressure of the CO_2 is reduced to now include a temperature/pressure combination for the CO_2 represented by point B*, the temperature/pressure combination at point B* is above the vaporization curve, and therefore the CO_2 is a low density fluid. As such, in moving from point B to point B* the CO_2 will have transitioned from a high density fluid to a low density phase state.

[0046] At point C, the temperature/pressure combination is above the vaporization curve represented by graphical line **204**, and therefore CO_2 at this temperature/pressure combination is in a low density fluid state. If the pressure of the CO_2 is reduced to now include a temperature/pressure combination for the CO_2 represented by point C*, point C* is still above the vaporization curve, and therefore CO_2 at the temperature/pressure combination represented by point C* is still in a low density fluid state, and has not changed phase state relative to CO_2 at point C due to the lowering of the temperature represented by the change in the temperature/pressure combination from point C to point C.

[0047] Thus, as illustratively represented by the points A, B, C, and A*, B*, and C* in phase diagram **200**, any phase-based control of the fluid using phase control flow valves cannot be based solely on temperature of the fluid, since at a same temperature of the CO_2 the CO_2 could be in a low density or a high density state. Further, the phase-based control of the fluid using these same phase control valves cannot be based solely on the pressure of the CO_2, since at a same pressure the CO_2 could be in a low density or a high density state. In the embodiments as described herein, it is desirable to have the CO_2 in a high-density fluid phase.

[0048] As further described below, the working fluid included in the sealed working fluid chamber of the flow control valve may be designed to have a vaporization curve that is the same as or similar to the vaporization curve for CO_2. As such, the working fluid will transition from a high density phase to the low density phase, and thus expand in volume, at a lower pressure then the fluid outside of the chamber. As such, when the CO_2 that is to be injected into the formation in a high density phase state, such as at point B in diagram **200**, is provided to the flow control valve, the working fluid in the valve may be at the lower pressure represented by point B*, and thus has transitioned to a low density phase state, thereby expanding and actuating the valve to a "closed" configuration. In the closed configuration, the flow of CO_2 through or past the flow control valve is blocked. As such, in order for the CO_2 that is intended to be injected into the formation to get past the flow control valve it must be provided to the flow control valve in not only the desired phase state, e.g., a high density fluid phase state, but also at a temperature/pressure combination that is far away enough from the vaporization curve so that the lower pressure of the working fluid contained within the sealed working fluid chamber of the flow control valve(s) does not transition to the low density phase state, but instead remains in the high density fluid phase state in order to maintain the flow control valve in the "open" configuration.

[0049] The injection of the CO_2 in the high-density fluid state allows for better control and a more even distribution of the CO_2 into the formation material. As such, embodiments of the flow control valve as described herein are configured to allow a flow of CO_2 past the flow control valve and to be injected into a subterrain formation when the CO_2 presented at the flow control valve is in the high density fluid phase state, and to block the flow of CO_2 into the subterranean formation when the CO_2 presented at the flow control valve is not in the high density fluid phase state, but instead is for example in the low density fluid phase state.

[0050] FIG. **2**B illustrates a phase diagram **220** for H_2. Phase diagram **220** includes a vertical axis **221** representing pressure in megapascal (Mpa), and horizontal axis **222** representing temperature in degrees Kelvin (K). Graphical line **224** represents a sublimation curve extending from the horizontal axis **222** to triple point **226**. Graphical line **228** represents a saturation curve and extends from triple point **226** to critical point **230**, and graphical line **232** represents the melting curve that extends upward and away from triple point **226**. The area **229**, which is bounded by an area to the right of sublimation curve **224** and saturation curve **228** and below dashed line **236**, includes temperature/pressure combinations where H_2 is in a gas phase state. The area **225**, which is bounded by and below melting curve **232**, above and bounded by saturation curve **228**, and bounded by and to the left of dashed line **234**, includes temperature/pressure combinations where H_2 is a liquid. The area **223**, which is above and to the left of saturation curve **224** and melting curve **232** includes temperature/pressure combinations where H_2 is in a solid phase. The area **227** below melting curve **232**, bounded on the left by dashed line **234**, and above and bounded by dashed line **236** includes temperature/pressure combinations where H_2 is in a supercritical state. In various embodiments, the H_2 that is to be delivered to the vapor-transition flow control valves for storge in a subterranean formation or cavity is to be in a supercritical phase state or liquid (high density fluid state), and the flow control valves are configured to actuate to an "open" configuration and allow a flow of the H_2 through the valves only when the H_2 presented to the valves is in the high density fluid state.

[0051] As such, embodiments of the vapor-transition flow control valves as described herein are configured to allow a flow of H_2 through or past the flow control valve, and to be injected into a subterrain formation or cavity when the H_2 presented at the flow control valve is in the high density fluid phase state, and to block the flow of H_2 into the subterranean formation or cavity when the H_2 presented at the flow control valve is not in the high density fluid phase state, for example in the gas or vapor phase states.

[0052] Given that disclosed above in FIGS. **2**A and **2**B, a high density fluid and a low density fluid may be defined in one or more different ways. In at least one embodiment, low density is less than the density of the fluid at its Triple Point (e.g., 467 kg/m^3 for CO_2) and high-density is anything greater than this point. In yet another embodiment, low density is the fluid in its gaseous phase while high density is the fluid in a non-gaseous phase e.g., (liquid or super critical). In even yet another embodiment, low and high density may be defined differently for different fluids. For example, for CO_2, low density would be a fluid with a density less than 200 kg/m^3, and high density would be a fluid with a density greater than this. Similarly, for H_2, low density would be a fluid with a density less than 10 kg/m^3, and high density would be a fluid with a density greater than this. In yet another embodiment, it could be defined by a ratio of the above.

[0053] Turning to FIG. **3**, illustrated is a cross-sectional view of a flow control valve **300** designed, manufactured and/or operated according to one or more embodiments of the disclosure. The flow control valve **300**, in one or more embodiments, includes a tubular **310**. The tubular **310**, in one embodiment, has a central longitudinal axis **315**, and is configured to be positioned downhole in a wellbore. For example, in at least one embodiment, the tubular **310** is configured to be connected in-line with a wellbore tubular (e.g., production tubular, coiled tubing, etc.).

[0054] The flow control valve **300**, in one or more other embodiments, further includes a piston **320** located in the tubular **310**. In one or more embodiments, the piston **320** is configured to separate injection fluid (e.g., not shown in FIG. **3**) located within the tubular **310** from a working fluid **340** located in a sealed working fluid chamber **330**, for example using one or more sealing members **322** (e.g., O-rings). The working fluid **340** may comprise any of the different working fluids disclosed above, for example depending on the operation of the device and the injection fluid used. Thus, in at least one embodiment, the working fluid **340** may comprise the same fluid as the injection fluid, such as CO_2, or an azeotrope of the injection fluid, such as an azeotrope of CO_2, among others.

[0055] The flow control valve **300**, in one or more embodiments, further includes a control arm **325** coupling the piston **320** to a valve assembly **350** located within the tubular **310**. In the illustrated embodiment of FIG. **3**, the control arm **325** forms at least a portion of a sliding sleeve. The valve assembly **350** may comprise a variety of different valve assemblies while remaining within the scope of the present disclosure. In one or more embodiments, however, the valve assembly **350** is a ball valve **360** having a central fluid aperture **365**. The central fluid aperture **365**, in one or more embodiments, is configured to be substantially parallel with the longitudinal axis **315** when the valve assembly **350** is in the open position and substantially transverse to the longitudinal axis **315** when the valve assembly **350** is in the closed position. The phrase "substantially parallel", when discussed with regard to the central fluid aperture **365**, means that the central fluid aperture **365** is within 10 degrees from parallel with the longitudinal axis **315**. The phrase "substantially transverse", when discussed with regard to the central fluid aperture **365**, means that the central fluid aperture **365** is within 10 degrees from transverse with the longitudinal axis **315**.

[0056] Turning to FIGS. **4**A through **4**F, illustrated are different operational states of a flow control valve **400** designed, manufactured and/or operated according to one or more embodiments of the disclosure. The flow control valve **400** of FIGS. **4**A through **4**F is similar in many respects to the flow control valve **300** of FIG. **3**. Accordingly, like reference numbers have been used to illustrate similar, if not identical, features.

[0057] With initial reference to FIG. **4**A, the flow control valve **400** is illustrated in its uphole state. Accordingly, the working fluid **340** within the sealed working fluid chamber **330** (e.g., without any injection fluid within the tubular **310** and/or wellbore fluid within the tubular **310**), holds the piston **320**, and thus the control arm **325**, toward the left (e.g., in this embodiment). Furthermore, with the piston **320** and the control arm **325** in the left most position, the valve assembly **350** is held in its closed position. In the illustrated embodiment of FIG. **4**A, the working fluid **340** has a working fluid pressure (P_{W0}) and a working fluid density (d_{W0}). In one example embodiment, the working fluid pressure (P_{W0}) and the working fluid density (d_{WO}) might be 780 psi and 150 kg/m^3, respectively.

[0058] Turning to FIG. **4**B, illustrated is the flow control valve **400** of FIG. **4**A after positioning it within a wellbore, but before pumping injection fluid down the tubular **310** and into engagement with the piston **320**. In the illustrated embodiment of FIG. **4**B, wellbore fluid **410** is located within the tubular **310** and imparting pressure on an opposite side of the piston **320** as the working fluid **340**. In the illustrated embodiment of FIG. **4**B, the wellbore fluid has **410** has a wellbore fluid pressure (P_1) and a wellbore fluid density (d_1), and the working fluid **340** has a working fluid pressure (P_{W1}) and a working fluid density (d_{W1}). In one example embodiment, the wellbore fluid pressure (P_1) and a wellbore fluid density (d_1) might be 15 psi and 1 kg/m^3, respectively, and the working fluid pressure (P_{W1}) and working fluid density (d_{W1}) might be 1200 psi and 150 kg/m^3, respectively, for example at a typical wellbore temperature of 200 degrees Fahrenheit.

[0059] Turning to FIG. **4**C, illustrated is the flow control valve **400** of FIG. **4**B after pumping injection fluid **420** down the tubular **310** and into engagement with the piston **320**. In the illustrated embodiment of FIG. **4**C, the injection fluid has **420** has an injection fluid pressure (P_{I2}) and an injection fluid density (d_{I2}), and the working fluid **340** still has the working fluid pressure (P_{W1}) and a working fluid density (d_{W1}). In one example embodiment, the injection fluid pressure (P_{I2}) and injection fluid density (d_{I2}) might be 1000 psi and 125 kg/m^3, respectively, and the working fluid pressure (P_{W1}) and a working fluid density (d_{W1}) might remain 1200 psi and 150 kg/m^3, respectively, again at a typical wellbore temperature of 200 degrees Fahrenheit. Given these example values, the piston **320** remains to the left (e.g., in the example embodiment).

[0060] Turning to FIG. **4**D, illustrated is the flow control valve **400** of FIG. **4**C after starting to increase the injection pressure of the injection fluid **420**. In the illustrated embodiment of FIG. **4**D, the injection fluid has **420** has an injection fluid pressure (P_{I3}) and an injection fluid density (d_{I3}), and the working fluid **340** has a working fluid pressure (P_{W3}) and a working fluid density (d_{W3}). In one example embodiment, the injection fluid pressure (P_{I3}) and injection fluid density (d_{I3}) might be 1600 psi and 225 kg/m^3, respectively, and the working fluid pressure (P_{W3}) and the working fluid density (d_{W3}) might be 1600 psi and 225 kg/m^3, respectively, again at a typical wellbore temperature of 200 degrees Fahrenheit. In this state, a volume of the sealed working fluid chamber **330** is only about 75 percent of the sealed working fluid chamber **330** of FIG. **4**C. As shown in FIG. **4**D, the valve assembly **350** is thus positioned about halfway between the open position and the closed position.

[0061] Turning to FIG. **4**E, illustrated is the flow control valve **400** of FIG. **4**D after continuing to increase the injection pressure of the injection fluid **420**. In the illustrated embodiment of FIG. **4**E, the injection fluid has **420** has an injection fluid pressure (P_{I4}) and an injection fluid density (d_{I4}), and the working fluid **340** has a working fluid pressure (P_{W4}) and a working fluid density (d_{W4}). In one example embodiment, the injection fluid pressure (P_{I4}) and injection fluid density (d_{I4}) might be 2000 psi and 300 kg/m^3, respectively, and the working fluid pressure (P_{W4}) and a working fluid density (d_{W4}) might be 2000 psi and 300 kg/m^3, respectively, again at a typical wellbore temperature of 200 degrees Fahrenheit. In this state, a volume of the sealed working fluid chamber **330** is about 50 percent of the sealed working fluid chamber **330** of FIG. **4**C. As shown in FIG. **4**E, the valve assembly **350** is now positioned in its open position.

[0062] Turning to FIG. **4**F, illustrated is the flow control valve **400** of FIG. **4**E after decreasing the injection pressure of the injection fluid **420**. In the illustrated embodiment of FIG. **4**F, the injection fluid has **420** has an injection fluid pressure (P_{I5}) and an injection fluid density (d_{I5}), and the working fluid **340** has a working fluid pressure (P_{W5}) and a working fluid density (d_{W5}). In one example embodiment, the injection fluid pressure (P_{I5}) and injection fluid density (d_{I5}) might be 1000 psi and 125 kg/m^3, respectively, and the working fluid pressure (P_{W5}) and a working fluid density (d_{W5}) might be 1200 psi and 150 kg/m^3, respectively, again at a typical wellbore temperature of 200 degrees Fahrenheit. In this state, a volume of the sealed working fluid chamber **330** is back to 100 percent of the sealed working fluid chamber **330** of FIG. **4**C. As shown in FIG. **4**F, the valve assembly **350** is now positioned back in its closed position.

[0063] It should be noted that in one or more embodiments, a pressure in the working fluid is coupled to an injection fluid pressure on an interior of the tubing. Similarly, in one or more embodiments, a temperature of the working fluid may be coupled with an injection fluid temperature.

[0064] Turning to FIG. **5**, illustrated is a cross-sectional view of a flow control valve **500** designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure. The flow control valve **500** of FIG. **5** is similar in many respects to the flow control valve **300** of FIG. **3**. Accordingly, like reference numbers have been used to indicate similar, if not identical, features. The flow control valve **500** of FIG. **5** differs, for the most part, from the flow control valve **300** of FIG. **3**, in that the flow control valve **500** includes a retention mechanism **510**, the retention mechanism **510** configured to require at least a minimum amount of force to release. The retention mechanism **510**, in one or more embodiments, is configured to prevent the valve assembly **350** from getting stuck in a half open position (e.g., as shown in FIG. **4**D), among other benefits.

[0065] In one or more embodiments, the retention mechanism **510** is configured to engage with the tubular **310**. For example, in one or more embodiments, the retention mechanism **510** is a control arm profile **520** extending (e.g., outwardly) from the control arm **325**. In this embodiment, the control arm profile **520** is configured to engage with one or more tubular profiles **530** of the tubular **310**, for example to require the at least the minimum amount of force to release. In the embodiment of FIG. **5**, the flow control valve **500** includes two tubular profiles **530***a*, **530***b*. In this embodiment, the retention mechanism **510** is configured to engage with the first tubular profile **530***a* of the tubular **310** to require at least a minimum amount of force to move the valve assembly **350** toward the open position, and is configured to engage with the second tubular profile **530***b* of the tubular **310** to require at least a minimum amount of force to move the valve assembly **350** toward the closed position. In the embodiment of FIG. **5**, the retention mechanism **510** is a collet. Nevertheless, those skilled in the art understand the myriad of different types of retention mechanisms that could be used and remain within the scope of the disclosure.

[0066] Turning to FIGS. **6**A through **6**I, illustrated are different operational states of a flow control valve **600** designed, manufactured and/or operated according to one or more embodiments of the disclosure. The flow control valve **600** of FIGS. **6**A through **6**I is similar in many respects to the flow control valve **500** of FIG. **5**. Accordingly, like reference numbers have been used to illustrate similar, if not identical, features.

[0067] With initial reference to FIG. **6**A, the flow control valve **600** is illustrated in its uphole state. Accordingly, the working fluid **340** within the sealed working fluid chamber **330** (e.g., without any injection fluid within the tubular **310** and/or wellbore fluid within the tubular **310**), holds the piston **320**, and thus the control arm **325**, toward the left (e.g., in this embodiment). Furthermore, with the piston **320** and the control arm **325** in the left most position, the valve assembly **350** is held in its closed position. In the illustrated embodiment of FIG. **6**A, the working fluid **340** has a working fluid pressure (P_{W0}) and a working fluid density (d_{W0}). In one example embodiment, the working fluid pressure (P_{W0}) and the working fluid density (d_{WO}) might be 780 psi and 150 kg/m^3, respectively.

[0068] Turning to FIG. **6**B, illustrated is the flow control valve **600** of FIG. **6**A after positioning it within a wellbore, but before pumping injection fluid down the tubular **310** and into engagement with the piston **320**. In the illustrated embodiment of FIG. **6**B, wellbore fluid **410** is located within the tubular **310** and imparting pressure on an opposite side of the piston **320** as the working fluid **340**. In the illustrated embodiment of FIG. **6**B, the wellbore fluid has **410** has a wellbore fluid pressure (P_1) and a wellbore fluid density (d_1), and the working fluid **340** has the working fluid pressure (P_{W1}) and a working fluid density (d_{W1}). In one example embodiment, the wellbore fluid pressure (P_1) and a wellbore fluid density (d_1) might be 15 psi and 1 kg/m^3, respectively, and the working fluid pressure (P_{W1}) and working fluid density (d_{W1}) might be 1200 psi and 150 kg/m^3, respectively, for example at a typical wellbore temperature of 200 degrees Fahrenheit.

[0069] Turning to FIG. **6**C, illustrated is the flow control valve **600** of FIG. **6**B after pumping injection fluid **420** down the tubular **310** and into engagement with the piston **320**. In the illustrated embodiment of FIG. **6**C, the injection fluid has **420** has an injection fluid pressure (P_{I2}) and an injection fluid density (d_{I2}), and the working fluid **340** has the working fluid pressure (P_{W1}) and the working fluid density (d_{W1}). In one example embodiment, the injection fluid pressure (P_{I2}) and injection fluid density (d_{I2}) might be 1000 psi and 125 kg/m^3, respectively, and the working fluid pressure (P_{W1}) and a working fluid density (d_{W1}) might remain 1200 psi and 150 kg/m^3, respectively, again at a typical wellbore temperature of 200 degrees Fahrenheit. Given these example values, the piston **320** remains to the left (e.g., in the example embodiment), for example as a result of the greater values for the working fluid pressure (P_{W1}) and a working fluid density (d_{W1}) than the injection fluid pressure (P_{I2}) and an injection fluid density (d_{I2}).

[0070] Turning to FIG. **6**D, illustrated is the flow control valve **600** of FIG. **6**C after starting to increase the injection pressure of the injection fluid **420**. In the illustrated embodiment of FIG. **6**D, the injection fluid has **420** has an injection fluid pressure (P_{I3}) and an injection fluid density (d_{I3}), and the working fluid **340** has the working fluid pressure (P_{W1}) and a working fluid density (d_{W1}). In one example embodiment, the injection fluid pressure (P_{I3}) and injection fluid density (d_{I3}) might be 1500 psi and 206 kg/m^3, respectively, and the working fluid pressure (P_{W1}) and the working fluid density (d_{W1}) might remain 1200 psi and 150 kg/m^3, respectively, again at a typical wellbore temperature of 200 degrees Fahrenheit. Typically, the difference in pressure values would push the piston **320** to the right. However, the retention mechanism **520** has the minimum amount of force (e.g., force differential across the piston **320**) to move, and the difference between the injection fluid pressure (P_{I3}) and an injection fluid density (d_{I3}), and the working fluid pressure (P_{W1}) and a working fluid density (d_{W1}) is not yet enough to overcome the minimum amount of force of the retention mechanism **520**.

[0071] Turning to FIG. **6**E, illustrated is the flow control valve **600** of FIG. **6**D after continuing to increase the injection pressure of the injection fluid **420**. In the illustrated embodiment of FIG. **6**E, the injection fluid has **420** has an injection fluid pressure (P_{I4}) and an injection fluid density (d_{I4}), and the working fluid **340** has a working fluid pressure (P_{W4}) and a working fluid density (d_{W4}). In one example embodiment, the injection fluid pressure (P_{I4}) and injection fluid density (d_{I4}) might be 2000 psi and 300 kg/m^3, respectively, and the working fluid pressure (P^{W4}) and the working fluid density (d_{W4}) might be 2000 psi and 300 kg/m^3, respectively, again at a typical wellbore temperature of 200 degrees Fahrenheit. In this embodiment, a difference between the injection fluid pressure (P_{I4}) and an injection fluid density (d_{I4}) and the working fluid pressure (P_{W3}) and a working fluid density (d_{W3}), was enough to overcome the minimum amount of force of the retention mechanism **520**, and thus the piston **320** started to move to the right. In the illustrated embodiment, the piston **320** does not stop at this midpoint, but continues to move from the state of FIG. **6**E to the state of FIG. **6**F in one continuous movement. In the illustrated embodiment, a pressure differential of at least 800 psi was required to overcome the amount of force of the retention mechanism. Nevertheless, the retention mechanism **510** may be designed to hold and/or release at any desirable pressure differential.

[0072] Turning to FIG. **6**F, illustrated is the flow control valve **600** of FIG. **6**E after the piston **320** continues to move to the right. In the illustrated embodiment of FIG. **6**F, the injection fluid has **420** has the same injection fluid pressure (P_{I4}) and the same injection fluid density (d_{I4}), and the working fluid **340** has the same working fluid pressure (P_{W4}) and the same working fluid density (d_{W4}). In this state, a volume of the sealed working fluid chamber **330** is about 50 percent of the sealed working fluid chamber **330** of FIG. **6**D. As shown in FIG. **6**F, the valve assembly **350** is now positioned in its open position.

[0073] Turning to FIG. **6**G, illustrated is the flow control valve **400** of FIG. **6**F after decreasing the injection pressure of the injection fluid **420**. In the illustrated embodiment of FIG. **6**G, the injection fluid has **420** has an injection fluid pressure (P_{I5}) and an injection fluid density (d_{I5}), and the working fluid **340** has the same working fluid pressure (P_{W4}) and a working fluid density (d_{W4}). In one example embodiment, the injection fluid pressure (P_{I5}) and injection fluid density (d_{I5}) might be 1300 psi and 169 kg/m^3, respectively, and the working fluid pressure (P_{W5}) and a working fluid density (d_{W5}) would remain 2000 psi and 300 kg/m$_3$, respectively, again at a typical wellbore temperature of 200 degrees Fahrenheit. In this embodiment, a difference between the injection fluid pressure (P_{I5}) and an injection fluid density (d_{I5}) and the working fluid pressure (P_{W4}) and a working fluid density (d_{W4}), is not enough to overcome the minimum amount of force of the retention mechanism **520**, and thus the piston **320** remains to the right.

[0074] Turning to FIG. **6**H, illustrated is the flow control valve **600** of FIG. **6**G after continuing to decrease the injection pressure of the injection fluid **420**. In the illustrated embodiment of FIG. **6**H, the injection fluid has **420** has an injection fluid pressure (P_{I6}) and an injection fluid density (d_{I6}), and the working fluid **340** has a working fluid pressure (P_{W6}) and a working fluid density (d_{W6}). In one example embodiment, the injection fluid pressure (P_{I6}) and injection fluid density (d_{I6}) might be 1200 psi and 150 kg/m^3, respectively, and the working fluid pressure (P_{W6}) and the working fluid density (d_{W6}) might be 1200 psi and 150 kg/m^3, respectively, again at a typical wellbore temperature of 200 degrees Fahrenheit. In this embodiment, a difference between the injection fluid pressure (P_{I6}) and an injection fluid density (d_{I6}) and the working fluid pressure (P_{W5}) and a working fluid density (d_{W5}), was enough to overcome the minimum amount of force of the retention mechanism **520**, and thus the piston **320** started to move to the left. In the illustrated embodiment, the piston **320** does not stop at this midpoint, but continues to move from the state of FIG. **6**H to the state of FIG. **6**I in one continuous movement.

[0075] Turning to FIG. **6**I, illustrated is the flow control valve **600** of FIG. **6**H after the piston **320** continues to move to the left. In the illustrated embodiment of FIG. **6**I, the injection fluid has **420** has the same injection fluid pressure (P_{I6}) and the same injection fluid density (d_{I6}), and the working fluid **340** has the same working fluid pressure (P_{W6}) and the same working fluid density (d_{W6}). As shown in FIG. **6**I, the valve assembly **350** is now positioned back in its closed position.

[0076] The embodiment of FIGS. **6**A through **6**I has inherent advantages, as it is designed to toggle between full-open and full-closed, for example without a significant likelihood of remaining within a partially open/closed state. Stated another way, the design of the embodiment of FIGS. **6**A through **6**I is configured to not be stable in a partially open/closed state.

[0077] Those skilled in the art to which this application relates will appreciate that other and further additions, deletions, substitutions and modifications may be made to the described embodiments.

1. A flow control valve, comprising:

a tubular configured to be positioned downhole in a wellbore, the tubular having a central longitudinal axis;

a piston located in the tubular, the piston configured to separate injection fluid located within the tubular from a working fluid located in a sealed working fluid chamber;

a valve assembly located within the tubular and coupled with the piston, wherein the piston is configured to axially slide within the tubular to move the valve assembly between an open position that provides a fluid passageway for the injection fluid from the tubular into a subterranean formation and a closed position that closes the fluid passageway for the injection fluid from the tubular into the subterranean formation, based upon a density of the injection fluid.

2. The flow control valve as recited in claim **1**, wherein a control arm couples the piston and the valve assembly.

3. The flow control valve as recited in claim **2**, wherein the control arm forms at least a portion of a sliding sleeve.

4. The flow control valve as recited in claim **2**, wherein the control arm includes a retention mechanism that is configured to engage with the tubular, the retention mechanism configured to require at least a minimum amount of force to release.

5. The flow control valve as recited in claim **4**, wherein the retention mechanism is a control arm profile extending from the control arm, the control arm profile configured to engage with one or more tubular profiles to require the at least the minimum amount of force to release.

6. The flow control valve as recited in claim **4**, wherein the retention mechanism is configured to engage with a first feature of the tubular to require at least a minimum amount of force to move the valve assembly toward the open position, and is configured to engage with a second feature of the tubular to require at least a minimum amount of force to move the valve assembly toward the closed position.

7. The flow control valve as recited in claim **1**, wherein the valve assembly is a ball valve having a central fluid aperture, the central fluid aperture configured to be substantially parallel with the longitudinal axis when the valve assembly is in the open position and substantially transverse to the longitudinal axis when the valve assembly is in the closed position.

8. The flow control valve as recited in claim **1**, wherein the working fluid is a gaseous working fluid located within the sealed working fluid chamber.

9. The flow control valve as recited in claim **8**, wherein the gaseous working fluid is CO_2.

10. The flow control valve as recited in claim **8**, wherein the gaseous working fluid is an azeotrope of CO_2.

11. A well system, comprising:

a wellbore extending through one or more subterranean formations;

a wellbore tubular positioned within the wellbore; and a flow control valve coupled with the tubular, the flow control valve including:

a tubular configured to be positioned downhole in the wellbore, the tubular having a central longitudinal axis;

a piston located in the tubular, the piston configured to separate injection fluid located within the tubular from a working fluid located in a sealed working fluid chamber;

a valve assembly located within the tubular and coupled with the piston, wherein the piston is configured to axially slide within the tubular to move the valve assembly between an open position that provides a fluid passageway for the injection fluid from the tubular into a subterranean formation and a closed position that closes the fluid passageway for the injection fluid from the tubular into the subterranean formation, based upon a density of the injection fluid.

12. The well system as recited in claim **11**, wherein a control arm couples the piston and the valve assembly.

13. The well system as recited in claim **12**, wherein the control arm forms at least a portion of a sliding sleeve.

14. The well system as recited in claim **12**, wherein the control arm includes a retention mechanism that is configured to engage with the tubular, the retention mechanism configured to require at least a minimum amount of force to release.

15. The well system as recited in claim **14**, wherein the retention mechanism is a control arm profile extending from the control arm, the control arm profile configured to engage with one or more tubular profiles to require the at least the minimum amount of force to release.

16. The well system as recited in claim **14**, wherein the retention mechanism is configured to engage with a first feature of the tubular to require at least a minimum amount of force to move the valve assembly toward the open position, and is configured to engage with a second feature of the tubular to require at least a minimum amount of force to move the valve assembly toward the closed position.

17. The well system as recited in claim **11**, wherein the valve assembly is a ball valve having a central fluid aperture, the central fluid aperture configured to be substantially parallel with the longitudinal axis when the valve assembly is in the open position and substantially transverse to the longitudinal axis when the valve assembly is in the closed position.

18. The well system as recited in claim **11**, wherein the working fluid is a gaseous working fluid located within the sealed working fluid chamber.

19. The well system as recited in claim **18**, wherein the gaseous working fluid is CO_2.

20. The well system as recited in claim **18**, wherein the gaseous working fluid is an azeotrope of CO_2.

21. A method, comprising:

providing injection fluid to a wellbore tubular extending downhole into a wellbore extending through a subterranean formation; and

controlling a flow of the injection fluid between the wellbore tubular and the subterranean formation using a flow control valve, the flow control valve including:

a tubular having a central longitudinal axis;

a piston located in the tubular, the piston configured to separate the injection fluid located within the tubular from a working fluid located in a sealed working fluid chamber;

a valve assembly located within the tubular and coupled with the piston, wherein the piston is configured to axially slide within the tubular to move the valve assembly between an open position that provides a fluid passageway for the injection fluid from the tubular into the subterranean formation and a closed position that closes the fluid passageway for the injection fluid from the tubular into the subterranean formation, based upon a density of the injection fluid.

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